Hold On to Your Cash:

CEO Speech Uncertainty and Financing Conservatism

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Abstract

We test whether speech uncertainty conveyed by chief executive officers during quarterly earnings calls predicts firm cash holdings, using a sample of 112,968 calls covering 2,429 U.S. firms over fiscal years 2002–2018. CEO speech uncertainty is measured by applying a finance-specific uncertainty wordlist to the CEO-tagged subset of speakers in each call's prepared-remarks segment (UncPreCEO) and spontaneous Q&A segment (UncAnsCEO). The Q&A channel raises both contemporaneous and one-quarter-lead cash-to-assets in all twelve specifications of a four-step fixed-effects panel; the Presentation channel is null. The cash response concentrates in credit-constrained Unrated firms in a 2002–2016 ratings sub-sample. Construct-validation regressions show the CEO speech-uncertainty metric responds positively to a firm-level political-risk index and to news-based US and global macroeconomic policy-uncertainty indices. Outside-world reaction regressions show that 26-day post-call quoted bid-ask spreads widen contemporaneously with the Presentation channel and at a one-quarter lag with the Q&A channel. The systematic application of the CEO speaker partition—rather than the canonical pooled-manager aggregate—to firm-quarter financing and external-reaction outcomes is the principal measurement contribution.

Keywords: CEO speech uncertainty, earnings calls, precautionary cash holdings, financing conservatism, financial constraints, textual analysis, bid-ask spreads.

JEL classification: G14, G32, M41, D83.

1 Introduction

1.1 Motivation

Corporate earnings calls offer a quarterly window into managerial communication that is higher in frequency than the annual 10-K disclosure cycle and richer in source-identifiable speaker attribution: a single call admits separate prepared-remarks and Q&A segments and identifies who is speaking (CEO, CFO, other manager, analyst). The central question we ask is whether uncertainty conveyed by the chief executive officer specifically, in either segment of the call, predicts the firm's contemporaneous and one-quarter-lead cash holdings—the precautionary response formalized by Opler, Pinkowitz, Stulz, and Williamson (1999): firms facing greater uncertainty about future cash flows hold more cash as a liquidity buffer against future funding shocks. Minton and Wruck (2001) document a cross-sectional cluster of low-leverage, high-cash firms whose pattern they label *financial conservatism*; the present paper inherits the noun (*Financing Conservatism*) for its title, but the empirical test targets the cash margin of that posture, leaving the leverage margin to future work.

1.2 Gap

The finance-specific uncertainty word list of Loughran and McDonald (2011) has been deployed extensively at the annual 10-K disclosure frequency, and Dzielínski, Wagner, and Zeckhauser (2021) and Hassan, Hollander, van Lent, and Tahoun (2020) have extended textual uncertainty measurement to the higher-frequency earnings-call corpus at firm-quarter resolution. To our knowledge, however, no prior work has systematically applied the CEO-specific speaker partition—enabled by the structured speaker-metadata fields in the Capital IQ vendor—to test the precautionary cash response of Opler et al. (1999) at firm-quarter frequency, nor has prior work jointly examined the firm's own cash-financing margin and the outside-world reaction margin (post-call quoted bid-ask spread) of CEO speech uncertainty. The CEO partition matters substantively in our setting: the empirical decomposition reported in Appendix D shows that pooling CEO speech with non-CEO manager speech, as in the all-manager aggregation convention of Bushee, Gow, and Taylor (2018), masks the concentration of the cash-holdings response in the CEO Q&A segment.

1.3 Approach

The speech-uncertainty regressor is constructed by applying the Loughran and McDonald (2011) uncertainty word list to earnings-call transcripts in the segment-split convention of Dzieliński et al. (2021), who document that a single call carries systematically different uncertainty content in its prepared-remarks and Q&A segments. The aggregation architecture follows the all-manager pooling of Bushee et al. (2018); we restrict that pooling to the CEO-tagged subset of speakers using the Capital IQ structured speaker-metadata fields. The Q&A regressor UncAnsCEO counts uncertainty words in CEO Q&A utterances divided by total CEO Q&A word count, and the Presentation regressor UncPreCEO is defined analogously on the prepared-presentation segment. The rationale for the CEO-specific partition—rather than aggregating over the full manager pool—is documented in §2.3; in brief, the choice rests on the upper-echelons role of the chief executive officer as principal external-facing communicator of the firm and on the empirical concentration of the cash-holdings response in the CEO Q&A segment.

1.4 Empirical Setting and Findings

The sample comprises 112,968 earnings-call transcripts covering 2,429 unique firms over fiscal years 2002–2018, sourced from S&P Capital IQ and merged with Compustat quarterly fundamentals and CRSP daily security data; financial firms (SIC 6000–6999) and regulated utilities (SIC 4900–4999) are excluded throughout, following Bates, Kahle, and Stulz (2009). The main empirical analyses test two hypotheses pre-committed in §2.1: HC (the cash response, §3.2) and HFC (the financial-constraint moderation of HC, §3.3). For HC, UncAnsCEO is positive and statistically significant in all twelve specifications of suite H1 across a four-step fixed-effects ladder, while the Presentation channel UncPreCEO is null across the same panel. For HFC, estimated on the 2002–2016 sub-sample where Compustat credit-rating coverage is available, the UncAnsCEO_c × Unrated interaction is positive and significant in six of eight interaction cells, with all four lead-dependent-variable cells reaching $p < 0.10$; the precautionary cash response therefore concentrates in the credit-constrained Unrated firms of Faulkender and Petersen (2006). Two additional sets of analyses follow. The driver regressions (§4.1) test whether the CEO speech-uncertainty metric itself responds to independently measured exogenous uncertainty: the firm-level political-risk index of Hassan et al. (2020) loads both CEO channels at high precision, and the news-based macro indices of Baker, Bloom, and Davis (2016) and Davis (2016) load the Presentation channel preferentially. The outside-world reaction regressions (§4.2) test whether liquidity providers price CEO speech uncertainty into post-call quoted spreads: the 26-day post-call closing-quote relative bid-ask spread ($\text{Spread}_{25\text{D}}$) widens contemporaneously with the Presentation channel and at a one-quarter lag with the Q&A channel. The Q&A-versus-Presentation channel asymmetries documented across the cash, driver, and spread analyses are reported descriptively, consistent with the pre-commitment of §2.1 that channel-specific asymmetry is reported ex post rather than asserted ex ante.

1.5 Contributions

The paper makes three contributions to the literature on managerial uncertainty disclosure and corporate financing.

1. **Systematic CEO-partitioned application to firm-quarter financing.** The framework of Dzieliński et al. (2021) establishes that conference-call transcripts admit a CEO-specific uncertainty metric at the individual-speaker level, and the architecture of Bushee et al. (2018) establishes the canonical pooled-manager aggregation. The principal measurement-side contribution of this paper is the systematic application of the CEO-specific partition—rather than the pooled-manager aggregate—to firm-quarter cash-financing outcomes (HC, HFC) and to firm-quarter external-reaction outcomes (§4.2). The CEO-partitioned regressor isolates a cash-response concentration that the pooled-manager aggregate obscures, as documented in Appendix D.

2. **Multi-aggregation-level construct validation of the CEO speech-uncertainty metric.** The driver regressions of §4.1 provide construct-validation evidence that the CEO-partitioned

measure carries economically meaningful uncertainty content at firm, US-macro, and global-macro aggregation levels, responding positively to the political-risk index of Hassan et al. (2020) and the news-based policy-uncertainty indices of Baker et al. (2016) and Davis (2016). The validation strengthens the substantive interpretation of HC and HFC as identifying an uncertainty channel of CEO speech rather than a stylistic artifact of the wordlist or the segment-split.

3. **Outside-world reaction in market liquidity.** The post-call spread regressions of §4.2 document that liquidity providers price both CEO speech segments into post-call closing-quote relative bid-ask spreads, with the Presentation channel registering contemporaneously within the call quarter and the Q&A channel registering with a one-quarter lag. The outside-world reaction margin is theoretically distinct from the firm-side cash response of §3, and the two findings are presented as descriptively complementary rather than as a tested causal sequence linking firm financing to market microstructure.

1.6 Roadmap

The remainder of the paper is organized as follows. Section 2 states the conceptual framework (§2.2), the speech-uncertainty measurement architecture (§2.3), the precautionary-motive and financing-conservatism literature anchors for HC and HFC (§2.4), and the empirical design including the fixed-effects ladder and clustering conventions (§2.5). Section 3 reports the data, sample, variables, and specification (§3.1), the HC results (§3.2), and the HFC results (§3.3). Section 4 reports the construct-validation driver regressions (§4.1) and the outside-world spread reaction (§4.2). Section 5 synthesizes the channel asymmetries across the three empirical sections, discloses limitations, and outlines five extensions for future research.

2 Conceptual Framework and Empirical Strategy

2.1 Pre-Commitment Statement

This section front-loads the statistical conventions used throughout the paper so that the empirical results in §3 and §4 can be read with the inference rules held constant.

Two formal hypotheses are stated in §3: HC (Cash—precautionary liquidity buffer, §3.2) and HFC (Financial Constraint moderation of HC, §3.3). Both are tested one-tailed in the theoretically predicted positive direction on both CEO-specific speech-uncertainty regressors, UncAnsCEO and UncPreCEO. The one-tailed convention is justified by the sign restriction imposed by the precautionary motive of Opler et al. (1999) and by the dominant theoretical direction in the related precautionary-motive and financing-conservatism literature. One-tailed inference applies symmetrically to both regressors; the Q&A and Presentation segments of CEO speech are both hypothesized to carry precautionary uncertainty content, and any channel-specific asymmetry observed empirically is reported descriptively rather than asserted ex ante.

Two additional analyses in §4 are tested under the same one-tailed positive convention but are not formal hypotheses. The driver regressions of §4.1 invert the direction of the main-suite specifications to ask whether the CEO speech-uncertainty metric itself responds to independently measured exogenous uncertainty shocks; we report them as construct-validation evidence for the main regressor. The outside-world reaction regressions of §4.2 ask whether post-call bid-ask spreads widen with CEO speech uncertainty, consistent with the canonical information-asymmetry prediction that liquidity providers widen quoted spreads to compensate for perceived informed-trading risk. Neither set of regressions is used as an identification strategy for HC or HFC.

All specifications reported in the paper body appear in their corresponding tables in their entirety; no column is dropped between estimation and reporting. The four-step fixed-effects ladder applied within each suite (§2.5) is a within-regressor robustness check, not an independent replication—the twelve columns of the HC table, for example, are not twelve independent tests of HC. Multiple-testing adjustment, consistent with the conventions above, is handled at the per-hypothesis rather than per-cell level.

2.2 Conceptual Framework

A large literature documents that aggregate uncertainty influences corporate decisions on financing and investment. The leading aggregate measures—the news-based US Economic Policy Uncertainty index of Baker et al. (2016) and its global extension by Davis (2016), together with implied-volatility series such as the Cboe Volatility Index (VIX)—are sampled at monthly or daily frequency but vary uniformly across firms within a period. They identify the time-series of uncertainty but not its cross-sectional incidence at a given firm.

A complementary literature recovers firm-specific uncertainty content from the firm's own corporate disclosures. Annual 10-K filings have been a primary source for textual measurement at firm-year frequency, with the canonical word-counting tradition introduced by Loughran and McDonald (2011) for the negative and uncertainty word lists. Hassan et al. (2020) extend the approach to earnings-call transcripts to construct a firm-quarter political-risk index, demonstrating that call speech contains firm-quarter-specific uncertainty content distinguishable from aggregate political shocks. Dzielínski et al. (2021) apply the Loughran and McDonald (2011) uncertainty word list to the same earnings-call corpus and show that uncertainty content varies across firms, across quarters within a firm, and across the prepared-remarks and Q&A segments of a single call.

The natural unit of analysis for testing precautionary behavior at firm-quarter frequency is therefore the call-level speech-uncertainty measure: it is sampled quarterly, identifies the firm rather than the macroeconomy, and admits source-identifiable decomposition of speakers within a single call (CEO, CFO, other managers, analysts). The remainder of §2 documents the speech-uncertainty construction we use (§2.3), the precautionary-motive theoretical anchor for the main hypotheses (§2.4), and the empirical design (§2.5).

2.3 Speech Uncertainty Measurement

The speech-uncertainty measure combines three elements: (i) a wordlist-based textual measurement at speaker-utterance resolution, (ii) aggregation within a structured speaker pool, and (iii) systematic application of a CEO-specific partition to firm-quarter outcomes. The first element is adopted from Loughran and McDonald (2011) and the second from Bushee et al. (2018); Dzielínski et al. (2021) document CEO-level measurement at the individual-speaker level, but the systematic application of the CEO partition to firm-quarter financing and external-reaction outcomes is the measurement-side contribution of this paper.

2.3.1 Wordlist and Segment Split

We apply the finance-specific uncertainty word list of Loughran and McDonald (2011), which classifies terms signaling uncertainty about future firm outcomes. The application to conference-call transcripts follows Dzielínski et al. (2021), who document that the transcript of a single call admits a presentation-versus-Q&A segment split: the presentation segment consists of prepared remarks authored in advance and read aloud, while the Q&A segment consists of spontaneous answers to analyst questions. The two segments differ systematically in uncertainty content, and our measure preserves the distinction.

2.3.2 Speaker Aggregation

Earnings-call transcripts identify speakers by structured metadata fields in the Capital IQ data vendor. Bushee et al. (2018) establish the convention of aggregating uncertainty counts across all manager speakers on a given call, as documented in the Ian Gow replication code accompanying that paper. Our measurement applies the same pooling architecture but partitions the manager pool along the speaker-role dimension, computing separate aggregates for CEO utterances and for non-CEO manager utterances.

2.3.3 Why the CEO Partition

Three considerations motivate the CEO-specific focus. First, the upper-echelons framework of Hambrick and Mason (1984) treats the CEO as the firm's principal strategic decision-maker and, correspondingly, as the authoritative source of uncertainty disclosure external-facing audiences attend to. Second, Bertrand and Schoar (2003) document empirically that individual CEOs leave idiosyncratic imprints on firm policies, implying that CEO-specific speech content is not a noisy substitute for the manager-pool average but carries distinct signal. Third, our own empirical decomposition (Appendix D) shows that the cash-holdings response to call-level speech uncertainty localizes to the CEO Q&A segment; the pooled-manager measure that subsumes CEO and non-CEO speech masks this concentration. Together the three considerations motivate the CEO-focused partition adopted in §3.

2.3.4 Construction

For each earnings call c, the Q&A speech-uncertainty regressor is

$$\text{UncAnsCEO}_c = \frac{N^{\text{LM-unc}}_{c,\text{CEO,QA}}}{N^{\text{all}}_{c,\text{CEO,QA}}},$$

where the numerator counts LM-uncertainty words in the CEO Q&A utterances of call c and the denominator counts all words in the CEO Q&A utterances of call c. The regressor is expressed as a share in $[0, 1]$ and pooled-winsorized at the 1st and 99th percentiles. The Presentation analogue UncPreCEO_c replaces "Q&A" with "Presentation" in both numerator and denominator. Calls in which the CEO role is not identified in the structured speaker metadata are coded as missing for the two CEO regressors; non-coverage combines two cases we do not separate in the coding—calls in which no speaker is tagged with a CEO role, and a smaller residual of calls with unresolved speaker-type placeholders in the Capital IQ metadata.

2.3.5 Sample Coverage

Under the 2002–2018 sample filter and the CEO-role-present filter, the CEO-partitioned sample covers approximately 80–81% of the call panel, compared with approximately 95–96% coverage for the manager-pool aggregate over the same period. The §3 specifications are estimated on the smaller CEO-present sample; descriptive statistics for both CEO regressors appear in Table 10.

2.4 Precautionary Motive and Financing Conservatism

The §3 hypotheses HC and HFC test whether greater CEO speech uncertainty predicts higher firm-quarter cash holdings, with the effect concentrated among credit-constrained firms. The theoretical and measurement anchors for these predictions are drawn from four strands of the precautionary-cash literature.

2.4.1 Precautionary-Motive Theory

Opler et al. (1999) formalize the precautionary motive for corporate cash holdings and predict that firms facing higher cash-flow uncertainty hold more cash as a liquidity buffer against future funding shocks. This theoretical prediction directly motivates HC: if the earnings-call CEO-speech signal captures forward-looking uncertainty about the firm, it should be contemporaneously associated with a higher cash-to-assets ratio.

2.4.2 Empirical Secular Pattern

Bates et al. (2009) document the cross-sectional and time-series persistence of the cash–uncertainty relation in the post-war US sample and establish the linear cash-to-assets ratio (`cheq/atq`) as the standard empirical form of the precautionary dependent variable. We adopt the Bates et al. ((2009)) dependent-variable specification directly in H1 (Table 11), together with their recommendation to cluster standard errors at the firm level (extended to two-way clustering in §4.1 for macro-IV suites).

2.4.3 Financial-Conservatism Cluster Label

Minton and Wruck (2001) identify a cross-sectional cluster of firms that persistently hold low leverage and high cash; they label this pattern *financial conservatism*. The Minton and Wruck ((2001)) label provides the framing noun *Financing Conservatism* that appears in this study's title: cash accumulation and leverage avoidance are two manifestations of a single posture, and the CEO speech-uncertainty signal identifies the quarterly uncertainty input that plausibly drives that posture. Although the present paper tests only the cash margin (HC, HFC) and does not test a leverage margin, the Minton and Wruck cluster framing makes the study's scope explicit.

2.4.4 Financial-Constraint Moderator

Faulkender and Petersen (2006) introduce a binary classification of public-debt-market access in which firms holding a bond rating are considered to have access and firms without a rating are considered to be credit constrained. The HFC hypothesis predicts that the precautionary cash response concentrates where access is most limited. We extend the Faulkender and Petersen ((2006)) binary scheme to a three-tier classification by partitioning rated firms into Investment-Grade (IG) and Below-Investment-Grade (BelowIG) subcategories; the Unrated category retains its credit-constrained benchmark role. The main-display estimand is the cash response to CEO speech uncertainty in Unrated firms relative to rated firms (IG and BelowIG pooled), with IG as the reference category for the level dummy (§3.3). The BelowIG interaction is suppressed from the body table (null and without economic content) and reported in Appendix C.

2.5 Empirical Design

All regressions are estimated by panel OLS via the `linearmodels` library. The unit of observation is an earnings call, indexed by firm identifier (gvkey) and fiscal quarter. The main HC and HFC specifications treat each firm's call as a single observation and absorb either calendar-year or calendar-year-quarter fixed effects across the four-step fixed-effects ladder below; the §4.1 driver regressions use the same unit but absorb only the calendar-year portion of that ladder (omitting calendar-year-quarter combinations) to preserve the within-year monthly variation that identifies the macro-uncertainty drivers.

2.5.1 Fixed-Effects Ladder

Each main suite reports four fixed-effects combinations, providing within-regressor robustness across progressively tighter controls for cross-firm and cross-time confounders:

1. Industry (Fama-French 12) fixed effects + calendar-year fixed effects;

2. Firm fixed effects + calendar-year fixed effects;

3. Industry fixed effects + calendar-year-quarter fixed effects;

4. Firm fixed effects + calendar-year-quarter fixed effects.

Each dependent variable (contemporaneous and one-quarter-lead) crosses the four FE configurations, yielding eight cells per DV and twelve columns for HC (unconditional specifications only) or sixteen columns for HFC (which additionally reports interaction specifications per DV $\times$ FE pair).

2.5.2 Controls

The base control set comprises Leverage, lnAssets, TobinsQ, ROA, Capex, DivDummy, sCFO, and a unit-by-DV Lagged_DV. The extended control set adds SalesGrowth, RDSales, CashFlowAt, and DailyVola. All controls are winsorized at the 1st and 99th percentiles on the Main-sample universe. The Lagged_DV term is the one-quarter lag of whichever dependent variable appears in that column—e.g., current-quarter CashRatio for the one-quarter-lead HC specification. Although Leverage (dlt/atq) and the cash-holdings dependent variable (cheq/atq) share a scaling denominator, they do not share a numerator, so Leverage is retained in the base control set rather than excluded as a bad control; the empirical robustness of the CEO-speech-uncertainty coefficients to the inclusion of Leverage is documented in the HC and HFC results of §3.

2.5.3 Standard Errors

Standard errors are clustered at the firm for the HC and HFC regressions (§3.2, §3.3) and for the outside-world reaction regressions (§4.2), following Bates et al. (2009). For the driver regressions (§4.1), which use monthly macro instruments (Baker et al. (2016), Davis (2016)), standard errors are two-way clustered at the firm and at the calendar-year-quarter to absorb residual cross-sectional dependence within macro periods.

2.5.4 Inference Convention

All reported p-values in the body of §3 and §4 are one-tailed on the speech-uncertainty regressors, consistent with the pre-commitment of §2.1. Two-tailed p-values on controls and on rating-category level dummies are reported in the appendix tables where relevant.

3 Main Empirical Analyses

3.1 Data, Sample, Variables, and Specification

We construct a quarterly panel of 112,968 earnings-call transcripts covering 2,429 unique firms over fiscal years 2002–2018, sourced from S&P Capital IQ. Call-level speech metrics are merged with Compustat quarterly fundamentals and CRSP daily security data. Following Bates et al. (2009), we exclude financial firms (SIC 6000–6999) and regulated utilities (SIC 4900–4999); we refer to this filtered universe as the Main sample throughout.

The dependent variable is CashRatio (`cheq/atq`, the linear cash-to-assets form adopted by Bates et al., (2009)). Speech uncertainty is measured by two CEO-specific call-level regressors: UncAnsCEO (CEO Q&A uncertainty) and UncPreCEO (CEO presentation uncertainty). Both are constructed by applying the Dzieliński et al. (2021) uncertainty wordlist at the speaker-utterance level and aggregating within the CEO speaker pool; full construction details, including the rationale for the CEO partition, appear in §2. Base controls are Leverage, lnAssets, TobinsQ, ROA, Capex, DivDummy, sCFO, and a unit-by-DV Lagged DV. Extended controls add SalesGrowth, RD-Sales, CashFlowAt, and DailyVola.

We estimate PanelOLS specifications across a four-step fixed-effects ladder (Industry + Calendar Year, Firm + Calendar Year, Industry + Calendar Year-Quarter, Firm + Calendar Year-Quarter), with standard errors clustered at the firm level. Each suite reports both the contemporaneous and the one-quarter-lead dependent variable across this FE ladder; per-suite column counts and the interaction structure (where applicable) are stated in the corresponding tables. Inference for both speech-uncertainty regressors in this section follows the one-tailed positive convention pre-committed in §2.1.

3.2 Cash Holdings (HC)

The first hypothesis tests the precautionary motive for corporate cash holdings introduced by Opler et al. (1999), who predict that "one would expect firms with greater cash flow uncertainty to hold more cash" (Section 2.1, p. 8). Bates et al. (2009) update this prediction in their secular study of U.S. corporations, documenting empirical persistence of the cash–uncertainty relation through 2006. Minton and Wruck (2001) characterize the joint cross-sectional pattern: firms persistently in the bottom leverage quintile—which they label *financially conservative*—hold cash equal to "21% of total assets on average... almost three times that of the typical control firm" (Section 3.2.1, p. 11).

Hypothesis HC. *An increase in CEO speech uncertainty during an earnings call is associated with a higher contemporaneous cash-to-assets ratio at the firm.*

HC is pre-specified one-tailed positive on both speech regressors. We test it on suite H1 (Table 11). The regressand is the linear cash-to-assets ratio of Bates et al. (2009); both main regressors (UncAnsCEO, UncPreCEO) enter every specification; the four-step fixed-effects ladder of §3.1 produces twelve columns reporting both contemporaneous and one-quarter-lead cash holdings.

The Q&A channel (UncAnsCEO) is positive and statistically significant in all twelve specifications. The contemporaneous coefficient ranges from +0.0018 to +0.0031 (p-values 0.0013–0.0150, one-tailed); the one-quarter-lead coefficient ranges from +0.0023 to +0.0028 (p-values 0.0111–0.0803). Four of the six contemporaneous cells reach the one-percent threshold; three of the six lead cells reach the five-percent threshold. Sample sizes range from 59,459 to 65,148 firm-quarters across columns.

The Presentation channel (UncPreCEO) does not reach conventional significance in any of the twelve specifications. Point estimates range from 0.0000 to +0.0016; the smallest one-tailed p-value across the twelve columns is 0.151. By the pre-commitment of §2.1, we fail to reject the null on the Presentation channel at any column. Descriptively, the prepared-presentation segment does not measurably move firm cash holdings at quarterly frequency; the asymmetry between the two CEO channels, with the precautionary signal concentrated in the spontaneous Q&A segment, is itself a finding to which we return in §5.

The economic magnitude is modest but consistent. UncAnsCEO has a sample standard deviation of 0.39 percentage points. A one-standard-deviation increase in CEO Q&A uncertainty therefore corresponds to between +0.0007 and +0.0012 in cash-to-assets across the FE ladder, or approximately 0.4–0.7% of the sample-mean cash ratio of 17.1%. Across the FE ladder the point estimate attenuates from industry-level to firm-level fixed effects, but the sign and significance survive every step. The pattern aligns with the Opler et al. (1999) directional prediction at higher temporal resolution than the firm-year panels of prior work, and the cash-up posture identified here is consistent with the high-cash side of the Minton and Wruck (2001) financial-conservatism cluster. §3.3 examines whether this response varies with firm-level access to public debt markets.

3.3 Financial Constraint Moderation (HFC)

The second hypothesis examines whether the precautionary cash response of §3.2 concentrates among credit-constrained firms. Faulkender and Petersen (2006) introduce a binary classification of public-debt-market access in which "firms that have a debt rating are classified as having access" (Section 1.2, p. 48), and document that "firms without a bond rating use significantly less debt... [evidence] that they are credit constrained" (Section 3.1, p. 62). The precautionary motive predicts that the cash response should bind tightest where access to public debt markets is most limited.

We extend the FP binary scheme to a three-tier hierarchy by partitioning rated firms into investment-grade (IG) and below-investment-grade (BelowIG) categories; the Unrated category retains its credit-constrained role per FP. We report the IG-versus-Unrated contrast in the main table (Table 14); the BelowIG interaction term is constructed but suppressed from the main display because the underlying coefficient is null and the row adds no economic content. The full BelowIG row is reported in Appendix C.

Hypothesis HFC. *The contemporaneous and one-quarter-lead cash response to CEO speech uncertainty is larger for firms in the Unrated category than for rated firms (IG and BelowIG pooled).*

HFC is pre-specified one-tailed positive on both main-effect coefficients (UncAnsCEO_c, UncPreCEO_c) and on both interaction terms (IV_c × Unrated). The mean-centered IVs use the Main-sample mean. We estimate H1.2 on the 2002–2016 subsample where Compustat credit-rating coverage is available; the panel narrows to approximately 57,000 firm-quarters per cell. The sixteen-column structure crosses the four-step fixed-effects ladder against contemporaneous and one-quarter-lead cash holdings; within each (FE × DV) cell we report an unconditional specification (no interaction terms) and an interaction specification (both IV_c × Unrated terms). Because the BelowIG interaction is dropped, the main-IV coefficients inside the interaction specifications apply to the rated category jointly (IG ∪ BelowIG) rather than to IG alone.

In the unconditional specifications the Q&A channel is significant in every cell. The contemporaneous UncAnsCEO_c coefficient ranges from +0.0017 to +0.0030 (p-values 0.0022–0.0273, one-tailed); the one-quarter-lead coefficient ranges from +0.0021 to +0.0024 (p-values 0.0290–0.0959). UncPreCEO_c is null throughout, with all eight one-tailed p-values above any conventional significance threshold — the same Presentation-channel pattern as the unconstrained suite in §3.2.

The interaction specifications confirm HFC for the Q&A channel and disconfirm it for the Presentation channel. The UncAnsCEO_c × Unrated coefficient is positive and significant in six of eight cells at the ten-percent threshold and four of eight at the five-percent threshold; on the one-quarter-lead dependent variable specifically, all four cells are significant at $p < 0.10$ and three of four at $p < 0.05$. The lead-cash interaction coefficient ranges from +0.0040 to +0.0069. A one-standard-deviation increase in CEO Q&A uncertainty therefore corresponds to roughly +0.002 additional cash-to-assets at Unrated firms relative to rated firms (the IG ∪ BelowIG pool), on top of the rated-firm response, or approximately 1.2% of the sample-mean cash ratio in the moderated component alone. The Unrated-versus-IG gap differs from the Unrated-versus-rated-pooled estimand reported here only by the small fraction of rated firms that fall in the BelowIG category.

The UncPreCEO_c × Unrated interaction is marginal at the contemporaneous DV (two of four cells at $p < 0.10$, both in industry-fixed-effects specifications) and null on the lead DV (zero of four). By the pre-commitment of §2.1, we fail to reject the null on the Presentation-channel interaction at conventional levels in four of eight cells and pass it only marginally in two: the constraint moderation does not load on the Presentation segment.

The combined evidence supports HFC through the spontaneous Q&A channel: credit-constrained Unrated firms exhibit a stronger cash response to CEO Q&A uncertainty than do Investment-Grade firms, and the moderation sharpens at the one-quarter-lead horizon. The Presentation-channel asymmetry observed for HC carries through to HFC, reinforcing the interpretation of the spontaneous Q&A segment as the locus of the precautionary signal in CEO earnings-call speech.

4 Additional Analyses

4.1 Drivers of CEO Speech Uncertainty

The main analyses of §3 treat CEO speech uncertainty as the regressor. This subsection inverts the direction and asks whether the speech-uncertainty metric itself responds to exogenous uncertainty shocks, a construct-validation test for the main IV. If the metric captures economically meaningful uncertainty content rather than stylistic noise, it should covary positively with independently measured uncertainty at firm, US, and global levels of aggregation.

We estimate three reverse-direction regressions. The firm-level driver is PRisk, the political-risk count of Hassan et al. (2020), which flags bigrams in a firm's earnings-call transcript that lie within ten words of a risk or uncertainty synonym and overlap a political-topic dictionary; PRisk therefore varies at the firm-quarter level, and the associated regressions cluster standard errors at the firm. The US macro driver is the news-based Economic Policy Uncertainty index of Baker et al. (2016), matched to each call by calendar month; the global aggregation is the GDP-weighted Global EPU index of Davis (2016), likewise matched monthly. The macro regressions cluster standard errors two-way at the firm and calendar-quarter levels and absorb calendar-year fixed effects, so identification runs off within-year monthly variation in the driver. All three regressions are tested one-tailed in the theoretically predicted positive direction (exogenous uncertainty shocks raise measured speech uncertainty). Table 23 consolidates the three drivers against the two CEO speech-uncertainty channels (UncAnsCEO, UncPreCEO) under both industry and firm fixed effects.

The firm-level PRisk driver is strongly significant in every cell. Contemporaneous coefficients range from +0.00013 to +0.00030 per unit PRisk across the four CEO cells (Q&A and Presentation channels × industry and firm fixed effects), with t-statistics between 9.0 and 17.0 and sample sizes of 65,394 to 65,760 firm-quarters. The PRisk-on-speech mapping therefore holds at high statistical precision across both CEO channels.

The macro drivers exhibit a channel-asymmetric pattern. US EPU raises UncPreCEO by +0.021 to +0.024 per unit log-EPU (p-values 0.0008–0.0021, one-tailed); the Q&A coefficient is smaller, +0.012 to +0.014, and reaches conventional significance only under firm fixed effects ($p = 0.037$; the industry-FE cell is $p = 0.052$). Global EPU follows the same pattern at larger magnitudes: UncPreCEO loads at +0.027 (industry FE, $p = 0.0056$) to +0.031 (firm FE, $p = 0.0021$); UncAnsCEO loads at +0.018 (industry FE, $p = 0.051$) and +0.021 (firm FE, $p = 0.027$).

The channel asymmetry here is inverted relative to the cash regressions of §3: the Q&A channel dominates in §3.2 and §3.3, but the Presentation channel dominates in the macro-driver regressions. The firm-level PRisk measure, which is itself constructed at the firm-call level, loads both channels approximately equally. The pattern is descriptively consistent with prepared remarks carrying broad uncertainty content (macroeconomic, political) and spontaneous Q&A content responding more tightly to firm-specific conditions; a fuller interpretive treatment is deferred to §5. Comparable results for the NoCEO speech channels, which identify whether uncertainty content in non-CEO speech also responds to the three drivers, are reported in Appendix B.1.

4.2 Outside-World Reaction

The preceding subsections show that CEO speech uncertainty co-moves with firm-level cash holdings (§3.2, §3.3) and responds to exogenous uncertainty shocks (§4.1). The complementary question is whether agents external to the firm recognize CEO speech uncertainty as a signal and react to it. The canonical market channel is information asymmetry, which liquidity providers price into the bid-ask spread of the firm's stock: a firm whose CEO conveys more uncertainty in its earnings call should have wider subsequent bid-ask spreads as market-makers widen quotes against informed counterparties.

We test this through the daily closing-quote relative bid-ask spread averaged over the call day and the 25 following trading days, denoted $\text{Spread}_{25\text{D}}$. The daily spread uses the standard closing-quote relative formulation, $2(\text{ASK}_d - \text{BID}_d)/(\text{ASK}_d + \text{BID}_d)$, computed from CRSP daily bid and ask quotes; the call-quarter observation is the simple mean across the call day and the 25 following trading days (a 26-day window), pooled-winsorized at the 1st and 99th percentiles. We adopt this window—wider than the 3-day announcement window typically used in short-horizon event studies—to capture market reaction sustained over approximately one trading month, beyond announcement-day volatility. The estimation uses Suite H14c, which enters the four call-level uncertainty IVs (UncAnsCEO, UncPreCEO, UncAnsMgr, UncPreMgr) jointly; we retain the manager-pool IVs here, unlike the CEO-only specification of §3, because the post-call bid-ask spread is plausibly priced against both CEO-specific and broader manager-pool speech content, and partialling out the manager-pool channel isolates the CEO-specific contribution to the market's revision of information asymmetry. We report the CEO coefficients in the body and the manager-pool coefficients in Appendix B.2. Standard errors are clustered at the firm. Inference is one-tailed positive on each speech-uncertainty IV, consistent with the information-asymmetry prediction.

The Presentation channel (UncPreCEO) raises the contemporaneous post-call spread. Across the six contemporaneous specifications (two fixed-effect families × three control sets) the coefficient ranges from +0.04 to +0.43; three of the six cells reach $p < 0.10$ (one-tailed), two reach $p < 0.05$, and one reaches $p < 0.01$. The strongest cells are the base specifications under industry and firm fixed effects (+0.30 at $p = 0.021$ and +0.43 at $p = 0.009$ respectively). Sample sizes range from 44,092 to

63,899 firm-quarters across the six cells. On the one-quarter-lead dependent variable, the Presentation channel does not reach significance in any cell (coefficients −0.03 to −0.24, all negative); under the one-tailed positive convention, the Presentation-channel effect on the post-call spread is therefore confined to the contemporaneous call quarter.

The Q&A channel (UncAnsCEO) does not move the contemporaneous spread: zero of six cells reach significance and point estimates span −0.10 to +0.07. It does, however, raise the one-quarter-lead spread across every specification. The lead coefficient ranges from +0.18 to +0.44; five of the six cells are significant at $p < 0.10$ and two reach $p < 0.05$. The one non-significant lead cell misses by less than half a percentage point ($p = 0.100$) and is sign-aligned with the five significant cells. Sample sizes for the lead specifications range from 44,569 to 63,443 firm-quarters.

The two CEO channels therefore produce a temporally complementary pattern: prepared remarks widen the post-call spread within the same call quarter, while spontaneous Q&A remarks widen the subsequent call-quarter spread. The contemporaneous Presentation effect is descriptively consistent with immediate price-formation reaction to the prepared statement of uncertainty, and the lagged Q&A effect with slower market assimilation of analyst-question-conditional content; fuller interpretation is deferred to §5. The two channel asymmetries identified in this section—Presentation dominates macro-driver regressions in §4.1 and contemporaneous spread here, while Q&A dominates the cash-holdings regressions of §3.2 and §3.3 and the lagged spread here—are descriptively consistent with the two segments of the call carrying distinct, economically meaningful uncertainty content that propagates through different downstream margins.

5 Discussion and Conclusion

5.1 Summary of Findings

The Cash Holdings hypothesis (HC, §3.2) is supported on the spontaneous Q&A channel of CEO speech across all twelve specifications of suite H1, with contemporaneous and one-quarter-lead positive coefficients on UncAnsCEO that survive the four-step fixed-effects ladder; the Presentation channel of CEO speech (UncPreCEO) is null in the same panel. The Financial Constraint moderation hypothesis (HFC, §3.3) confirms HC for the Q&A channel and concentrates the precautionary cash response in Unrated firms relative to the rated pool: the UncAnsCEO_c × Unrated interaction is positive and significant in six of eight interaction cells, with all four lead-DV cells reaching $p < 0.10$. The driver regressions (§4.1) confirm that the speech-uncertainty metric responds positively to the firm-level PRisk index of Hassan et al. (2020) across both CEO channels, and to the news-based macro indices of Baker et al. (2016) and Davis (2016) preferentially through the Presentation channel. The outside-world reaction regressions (§4.2) show that the post-call 26-day closing-quote relative bid-ask spread ($\text{Spread}_{\text{25D}}$) widens contemporaneously with UncPreCEO and at a one-quarter lag with UncAnsCEO. The economic magnitude of the cash response is modest—a one-standard-deviation increase in UncAnsCEO corresponds to between 0.4% and 0.7% of the sample-mean cash-to-assets ratio across the FE ladder—but statistically robust, and the moderator specification confirms that UncPreCEO_c and its Unrated interaction are null or marginal across the same ladder.

5.2 Channel-Asymmetry Synthesis

Three empirical patterns identify a Q&A-versus-Presentation asymmetry across the analyses. First, the cash response of §3.2 and §3.3 loads on the spontaneous Q&A channel and not on the prepared Presentation channel. Second, the macro-uncertainty drivers of §4.1 (US EPU, Global EPU) load preferentially on the Presentation channel and load the Q&A channel only weakly under firm fixed effects. Third, the post-call spread response of §4.2 is contemporaneous on the Presentation channel and lagged on the Q&A channel. The firm-call-level PRisk driver of §4.1 is a boundary condition: it loads both CEO channels at approximately equal magnitude and at high precision, which is descriptively consistent with the channel asymmetry being a function of the aggregation level of the regressor (firm-quarter versus monthly macroeconomic) rather than of the segment alone. Per the pre-commitment of §2.1, we report this asymmetry descriptively and do not theorize a structural mechanism for the prepared-versus-spontaneous split beyond the observation that the two segments carry distinct uncertainty content propagating through different downstream margins.

5.3 Mechanism Disclosure

The findings of §3 and §4.2 operate on theoretically distinct margins. The cash-holdings results test the firm's financing margin: greater perceived future-funding uncertainty raises the precautionary liquidity buffer, consistent with the Opler et al. (1999) prediction extended to a within-firm panel resolution at quarterly frequency. The post-call spread results test a separate, market-side margin: liquidity providers widen quoted spreads when CEO speech raises perceived informed-trading risk, a canonical information-asymmetry mechanism distinct from the firm's own funding response. The present paper does not test a causal bridge linking the two margins—e.g., whether the spread widening drives the cash accumulation, or whether the cash accumulation calms a subsequent spread, or whether both respond independently to a common shock. The two findings are descriptively complementary; their joint identification is left to future work.

5.4 Limitations

Five caveats bound the interpretation. First, all analyses are observational. The speech-uncertainty regressors are not exogenously assigned, and the §3 and §4.2 estimates identify associations consistent with the precautionary and information-asymmetry mechanisms invoked above without identifying either causally. Second, CEO speaker identification depends on the Capital IQ structured speaker-metadata fields documented in §2.3; the CEO-partitioned sample covers approximately fifteen percentage points fewer calls than the manager-pool aggregate over the same period, with this coverage gap dominated by calls in which no speaker is tagged with a CEO role and a smaller residual of unresolved speaker-type placeholders. Third, the HC sample covers 2002–2018 and the HFC moderator analysis is bounded at 2016 by Compustat credit-rating coverage; HFC results in the 2017–2018 sub-period are therefore not tested. Fourth, the post-call spread window of §4.2 is a 26-day construction (the call day plus 25 following trading days) chosen to capture market reaction sustained beyond the announcement window; alternative window lengths are not estimated in the body. Fifth, the data pipeline retains the internal column name `BGTLevel_Spread` for backward compatibility with prior runs; the user-facing label Spread_{25D} is used consistently in the body and tables, but legacy specification JSON and panel artifacts reflect the prior nomenclature.

5.5 Future Research

Five extensions are flagged as natural follow-ups. First, an identification strategy for the speech-uncertainty effect on cash, exploiting plausible exogenous shocks to CEO communication style (e.g., disclosure-rule changes, exogenous CEO turnover, or natural experiments in mandatory-disclosure timing) to recover causal estimates against the observational baseline of §3. Second, a within-firm temporal-sequencing test linking the cash response of §3.2 and the spread response of §4.2 within a common firm-quarter window, to test whether the two margins are temporally ordered or operate in parallel. Third, additional cash-response moderators beyond the Unrated category of §3.3, including size quartiles, dividend-payer status, and continuous credit-constraint indices, to identify complementary constraint dimensions that the binary FP scheme aggregates. Fourth, an OPSW log-cash-DV robustness specification of HC that estimates the cash response under the alternative log-form dependent variable, complementing the linear Bates et al. (2009) specification adopted in the body. Fifth, full pipeline rename of the spread variable to Spread_{25D} across the underlying panel, builders, and specification JSON, removing the BGT-window legacy column name retained in this draft for backward compatibility.

5.6 Concluding Remark

The CEO partition of the call-level speech-uncertainty measure identifies a firm-quarter precautionary signal that the manager-pooled aggregate masks: the Q&A channel of CEO speech is positively associated with contemporaneous and one-quarter-lead cash holdings (HC), with the response sharpest at credit-constrained Unrated firms (HFC). Construct validation against firm-level (Hassan et al. (2020)) and macro-level (Baker et al. (2016), Davis (2016)) uncertainty drivers confirms that the metric carries economically meaningful uncertainty content across multiple aggregation levels. Outside-world reaction evidence shows that market liquidity providers price both CEO speech segments into post-call quoted spreads, with the Presentation channel registering contemporaneously and the Q&A channel registering with a one-quarter lag. The two channel asymmetries documented in the body—Q&A dominant for cash and lagged spread, Presentation dominant for macro-driven speech-content variation and contemporaneous spread—are descriptively consistent with the prepared and spontaneous segments of an earnings call carrying distinct, economically meaningful uncertainty content that propagates through different downstream margins. The systematic application of the CEO partition to firm-quarter financing and external-reaction outcomes is the principal measurement contribution of this paper, and motivates the extensions outlined above.

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A Variable Definitions

This appendix lists every variable used in the empirical analyses, with construction formula, data source, and originating reference. Variable names follow paper-standard conventions where available (Dzieliński et al. 2021 for speech measures, Bates et al. 2009 for cash-holdings controls); deviations from source-paper frequency or sample are documented in the formula column.

A.1 Sample Manifest

Name	Description / Formula	Source	Reference
`file_name`, `ceo_id`, `ceo_name`, `gvkey`, `ff12_code`, `ff12_name`, `start_date`	Master sample manifest with CEO, firm, and industry identifiers	outputs/1.4_AssembleManifest	

A.2 Text/Linguistic Variables

Name	Description / Formula	Source	Reference
`NegCall`	Entire-call negative sentiment percentage. *Formula:* Ratio of negative words to total words in the entire call, based on LM 2011 negative-words list. Per DWZ 2021 Appendix Table A.1: 'percentage of negative words in all words spoken by the CEO, CFO and analysts attending the call.'	outputs/2_Textual_Analysis/2.2_Variables	dzielinski2021 (Section 4.4, p.18; Appendix Table A.1, p.54)
`UncAnsCEO`	CEO-only Q&A uncertainty percentage. *Formula:* CEO Q&A uncertainty (%) = UnctWordsAnsCEO / WordsAnsCEO (DWZ 2021 Eqn 2). Uncertainty wordlist = LM 2011 Master Dictionary 'uncertainty' list (297 words, Aug 2014 version). CEO identified via Execucomp ceoann field.	outputs/2_Textual_Analysis/2.2_Variables	dzielinski2021 (Section 4.2, Eqn 2, p.13; Appendix Table A.1, p.54)
`UncAnsMgr`	Manager Q&A uncertainty percentage (all managers including CEO). *Formula:* Manager Q&A uncertainty (%) = (count of LM 2011 uncertainty-wordlist words spoken by managers in Q&A segment) / (total words spoken by managers in Q&A segment). Construction strategy follows DWZ 2021 Eqn (2); manager-pool identification follows BGT 2018 segment-split code (iangow/bgt replication). 'Manager' pool includes CEO plus other managerial speakers, extending DWZ which separates CEO/CFO.	outputs/2_Textual_Analysis/2.2_Variables	dzielinski2021 (measurement strategy, Eqn 2); bushee2018 (manager-pool identification via iangow/bgt); thesis extension to pooled manager speakers
`UncPreCEO`	CEO-only presentation uncertainty percentage. *Formula:* CEO Presentation uncertainty (%) = UnctWordsPreCEO / WordsPreCEO (DWZ 2021 Eqn 1). Uncertainty wordlist = LM 2011 (297 words).	outputs/2_Textual_Analysis/2.2_Variables	dzielinski2021 (Section 4.2, Eqn 1, p.13; Appendix Table A.1, p.54)
`UncPreMgr`	Manager presentation uncertainty percentage. *Formula:* Manager Presentation uncertainty (%) = (count of LM 2011 uncertainty-wordlist words by managers in Presentation segment) / (total manager words in Presentation segment). Construction strategy follows DWZ 2021 Eqn (1); manager-pool identification and Pres/Ans segment split follow BGT 2018 via iangow/bgt replication code.	outputs/2_Textual_Analysis/2.2_Variables	dzielinski2021 (measurement strategy, Eqn 1); bushee2018 (segment split + manager pool); thesis extension to pooled manager speakers
`UncQue`	Analyst Q&A uncertainty percentage. *Formula:* Analyst Q&A uncertainty (%) = UnctWordsQue / WordsQue (DWZ 2021 Eqn 3). Uncertainty wordlist = LM 2011 (297 words).	outputs/2_Textual_Analysis/2.2_Variables	dzielinski2021 (Section 4.2, Eqn 3, p.13; Appendix Table A.1, p.54)

A.3 Financial / Market Variables

Name	Description / Formula	Source	Reference
BGTAvg_Amihud	BGT 25-day symmetric average Amihud illiquidity (H7e). *Formula:* Mean daily Amihud illiquidity over [-25, +25] trading days around call (51-day symmetric window, day 0 INCLUDED). F1D extension to BGT 2018.	.../variables/bgt_long_window_amihud.py	bgt2018 (window); F1D extension (symmetric average shape)
BGTAvg_Spread	BGT 25-day symmetric average bid-ask spread (H14e). *Formula:* Mean daily relative bid-ask spread over [-25, +25] trading days around call (51-day symmetric window, day 0 INCLUDED). Spread = (ASK - BID) / ((ASK + BID) / 2). F1D extension to BGT 2018.	.../variables/bgt_long_window_spread.py	bgt2018 (window); F1D extension (symmetric average shape)
BGTDelta_Amihud	BGT 25-day post-pre Amihud illiquidity change (H7d). *Formula:* Mean Amihud over post window [+1, +25] minus mean over pre window [-25, -1] trading days (day 0 EXCLUDED). F1D extension to BGT 2018.	.../variables/bgt_long_window_amihud.py	bgt2018 (window); F1D extension (delta shape)
BGTDelta_Spread	BGT 25-day post-pre bid-ask spread change (H14d). *Formula:* Mean spread over [+1, +25] minus mean over [-25, -1] trading days (day 0 EXCLUDED). F1D extension to BGT 2018.	.../variables/bgt_long_window_spread.py	bgt2018 (window); F1D extension (delta shape)
BGTLevel_Amihud	Compute BGT (2018) long-window Amihud illiquidity around earnings calls. *Formula:* CRSP via get_raw_daily_data (see module: src/f1d/shared/variables/bgt_long_window_amihud.py)	CRSP via get_raw_daily_data	Bushee, Gow & Taylor (2018, JAR) 56(1):85-121
BGTLevel_Spread	Compute BGT (2018) long-window closing bid-ask spread around earnings calls. *Formula:* CRSP via get_raw_daily_data (closing BID/ASK) (see module: src/f1d/shared/variables/bgt_long_window_spread.py)	CRSP via get_raw_daily_data	Bushee, Gow & Taylor (2018, JAR) [window]; Lee (2016, TAR) [formula]
Capex	Build Capex = capxy_annual / atq_lag from raw Compustat quarterly data. *Formula:* Compustat/capxy_Q4/atq (see module: src/f1d/shared/variables/capex_intensity.py)	Compustat/capxy_Q4/atq	Bates, Kahle & Stulz (2009, JF)
CashFlowAt	Build CashFlow = oancfy / atq from raw Compustat quarterly data. *Formula:* Compustat/oancfy/atq (see module: src/f1d/shared/variables/cash_flow.py)	Compustat/oancfy/atq	Bates, Kahle & Stulz (2009, JF)
CashRatio	Build CashRatio = cheq / atq from raw Compustat quarterly data. *Formula:* Compustat/cheq/atq (see module: src/f1d/shared/variables/cash_holdings.py)	Compustat/cheq/atq	bates2009 (Section II, p.1991; BKS use annual che/at — we use quarterly cheq/atq for panel alignment)
ChangDebtChoice	ChangDebtChoice (description not in module docstring) *Formula:* see module: src/f1d/shared/variables/_compustat_engine.py	.../variables/_compustat_engine.py	cdh2006
ChangExternalFunding	ChangExternalFunding (description not in module docstring) *Formula:* see module: src/f1d/shared/variables/_compustat_engine.py	.../variables/_compustat_engine.py	cdh2006
DailyVola	Build Volatility from raw CRSP daily stock files via CRSPEngine. *Formula:* CRSP/RET (see module: src/f1d/shared/variables/volatility.py)	CRSP/RET	dwz2021
DebtToCapital	Build DebtToCapital = total debt / (equity + total debt). *Formula:* Compustat/dlcq,dlttq,seqq (see module: src/f1d/shared/variables/debt_to_capital.py)	Compustat/dlcq,dlttq,seqq	rz1995
DivDummy	Build DivDummy = (dvy ¿ 0).astype(float) from raw Compustat quarterly data. *Formula:* Compustat/dvy¿0 (see module: src/f1d/shared/variables/dividend_payer.py)	Compustat/dvy¿0	bates2009 (Section IV, p.2000; dividend payout dummy based on common dividends)

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Name	Description / Formula	Source	Reference
DivPayerQ	Build DivPayerQ = (dvpsxq ¿ 0).astype(float) from raw Compustat quarterly data. *Formula:* Compustat/dvpsxq¿0 (quarterly ex-date) (see module: src/f1d/shared/variables/dividend_payer_quarterly.py)	Compustat/dvpsxq¿0	Hoberg & Prabhala (2009, RFS) — Compustat item 26 (dvpsx) analog
EarnVol	Builder for Earnings Volatility (earnings_volatility) variable. *Formula:* Compustat (see module: src/f1d/shared/variables/earnings_volatility.py)	Compustat	duong2025 (Appendix B, p.30; 3-year rolling std of qoq asset-scaled earnings)
EquityDelayCon	Hoberg-Maksimovic (2015) equity-market financing constraint index (H22). *Formula:* Hoberg-Maksimovic (2015) firm-year equity-market financing constraint measure. Higher values = more constrained firm. Panel column is lowercase ‘equitydelaycon‘; CamelCase name is the spec convention. See docs/provenance/h22.md for merge details.	inputs/Hoberg_Maksimovic/	hm2015
FirmMat	Builder for Firm Maturity (firm_maturity) variable. *Formula:* Compustat (see module: src/f1d/shared/variables/firm_maturity.py)	Compustat	biddle2009 (Appendix A, p.130; years since first CRSP appearance); alt leary2010 (App.˜A p.354)
Leverage	Build Leverage = (dlcq + dlttq) / atq (interest-bearing debt only). *Formula:* Compustat/dlcq,dlttq,atq (see module: src/f1d/shared/variables/book_lev.py)	Compustat/dlcq,dlttq,atq	bks2009, rz1995
lnAssets	Build Size = ln(atq) from raw Compustat quarterly data. *Formula:* Compustat/atq (see module: src/f1d/shared/variables/size.py)	Compustat/atq	dwz2021
Loss	Builder for Loss (H5 Control Variable). *Formula:* Compustat ibq ¡ 0 (see module: src/f1d/shared/variables/loss_dummy.py)	Compustat ibq ¡ 0	biddle2009 (Appendix A, p.131; indicator for negative net income before extraordinary items)
PayoutRatio_q	Build PayoutRatio_q = (dvpspq × cshoq) / ibq from Compustat. *Formula:* Compustat/(dvpspq*cshoq)/ibq (quarterly) (see module: src/f1d/shared/variables/payout_ratio_quarterly.py)	Compustat/	dds2004
PRisk	Match quarterly PRisk onto each earnings call by (gvkey, calendar_quarter). *Formula:* Hassan et al. (2019) quarterly PRisk (see module: src/f1d/shared/variables/prisk_q.py)	Hassan et al.	hhlt2019
RDSales	Build RDSales = xrdq / atq from raw Compustat quarterly data. *Formula:* Compustat/xrdq,atq (see module: src/f1d/shared/variables/rd_intensity.py)	Compustat/xrdq,atq	bks2009, jjl2021
RepurchaseIntensity	Build RepurchaseIntensity = quarterly_prstkcy / lagged_atq from Compustat. *Formula:* Compustat/quarterly_prstkcy/lagged_atq (see module: src/f1d/shared/variables/repurchase_intensity.py)	Compustat/quarterly_prstkcy/lagged_atq	Thesis Advisor Suggestion
ROA	Build ROA = iby_annual / avg_assets from Compustat quarterly data. *Formula:* Compustat/iby,atq (see module: src/f1d/shared/variables/roa.py)	Compustat/iby,atq	wang2020, dwz2021
SalesGrowth	Build SalesGrowth from raw Compustat quarterly data (annual Q4 panel). *Formula:* Compustat/saley_Q4_YoY (see module: src/f1d/shared/variables/sales_growth.py)	Compustat/saley_Q4_YoY	biddle2009 (Section 3.2, p.117; pct change in sales)
sCFO	Build sCFO = rolling 5-year std (min 3 yrs) of (oancfy/atq_{t-1}) per gvkey. *Formula:* Compustat/oancfy/atq_{t-1} rolling std (see module: src/f1d/shared/variables/ocf_volatility.py)	Compustat/oancfy/atq_{t-1} rolling std	biddle2009 (Appendix A, p.130; 5-year rolling std of CFO/avg assets)

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Name	Description / Formula	Source	Reference
`StockPrice`	Compute stock price at earnings call date (VECTORIZED). *Formula:* CRSP via get_raw_daily_data (see module: src/f1d/shared/variables/stock_price.py)	CRSP via get_raw_daily_data	amiram2016 (Appendix A, p.137; daily CRSP stock price; justification cites Stoll 1978)
`SurpDec`	Signed-quintile earnings surprise rank (-5 to +5) within calendar quarter. *Formula:* Signed quintile rank of percentage earnings surprise per DWZ 2021 Appendix Table A.1: raw surprise = (actual - consensus forecast earnings) / share price 5 trading days before announcement, x 100. Firms grouped into 5 bins of positive surprise (+5 largest through +1 smallest positive), 0 for zero surprises, and 5 bins of negative surprise (-1 smallest negative through -5 largest negative). DWZ footnote 32 notes this decile convention follows Hirshleifer, Lim & Teoh (2009) and DellaVigna & Pollet (2009).	.../variables/earnings_surprise.py	dzielinski2021 (Appendix Table A.1, p.55; convention cites DellaVigna-Pollet 2009 + Hirshleifer-Lim-Teoh 2009)
`TobinsQ`	Build TobinsQ = (AT + cshoq*prccq - CEQ) / AT from raw Compustat quarterly data. *Formula:* Compustat/(atq+cshoq*prccq-ceqq)/atq (see module: src/f1d/shared/variables/tobins_q.py)	Compustat/	Bates, Kahle & Stulz (2009, JF)
`Turnover`	Compute share turnover at earnings call date (VECTORIZED). *Formula:* CRSP via get_raw_daily_data (see module: src/f1d/shared/variables/turnover.py)	CRSP via get_raw_daily_data	amihud2002 (Section 2.1, p.33; alt rolling-window def in chang2006 Appendix B p.3045)

A.4 Econometric / Indicator Variables

Name	Description / Formula	Source	Reference
`BelowIG`	Below-IG credit rating subsample indicator (H1.2). *Formula:* Binary = 1 if firm's S&P long-term issuer credit rating is below investment grade (BB+ and below) at call date; 0 otherwise. Rated firms only (Unrated firms get separate indicator).	runtime	thesis (H1.2 rating subsample)
`HighTSIMM`	High product-market-similarity indicator (H1.1). *Formula:* Binary = 1 if firm-year TotalSimilarity (HP 2016 product market similarity) is above the within-year median; 0 otherwise.	runtime	hp2016 (TSIMM construction); thesis (H1.1 split convention)
`Lagged_DV`	Generic lagged-DV control (1Q lag of suite-specific DV). *Formula:* 1-quarter lag of the dependent variable, constructed at runtime per spec (DV varies per suite). Always included as base control to absorb persistence.	runtime	thesis convention (per feedback_lagged_dv.md)
`Unrated`	No-rating subsample indicator (H1.2). *Formula:* Binary = 1 if firm has no S&P rating at call date; 0 otherwise.	runtime	thesis (H1.2 rating subsample)
`z_log_TotalSimilarity`	z-scored log of HP 2016 TotalSimilarity (continuous moderator, H1.1). *Formula:* z-score of log(1 + TotalSimilarity) computed within year. TotalSimilarity = HP 2016 firm-year product market similarity score (sum of pairwise cosine similarities $\times$ 100).	runtime	hp2016

A.5 Runtime Derivatives (Lead/Lag/Centered/Interaction)

Name	Description / Formula	Source	Reference
AbsSurpDec	Absolute earnings surprise quintile magnitude. *Formula:* abs(SurpDec) — unsigned magnitude of signed-quintile earnings surprise rank, values 0-5.	runtime	thesis (magnitude control independent of sign)
BGTAvg_Amihud_lead1	1Q lead of BGTAvg_Amihud. *Formula:* 1-quarter lead of BGTAvg_Amihud	runtime	bgt2018 (window); F1D extension (symmetric average shape)
BGTAvg_Spread_lead1	1Q lead of BGTAvg_Spread. *Formula:* 1-quarter lead of BGTAvg_Spread	runtime	bgt2018 (window); F1D extension (symmetric average shape)
BGTDelta_Amihud_lead1	1Q lead of BGTDelta_Amihud. *Formula:* 1-quarter lead of BGTDelta_Amihud	runtime	bgt2018 (window); F1D extension (delta shape)
BGTDelta_Spread_lead1	1Q lead of BGTDelta_Spread. *Formula:* 1-quarter lead of BGTDelta_Spread	runtime	bgt2018 (window); F1D extension (delta shape)
BGTLevel_Amihud_lead1	1Q lead of BGTLevel_Amihud. *Formula:* 1-quarter lead of BGTLevel_Amihud	runtime	bgt2018
BGTLevel_Spread_lead1	1Q lead of BGTLevel_Spread. *Formula:* 1-quarter lead of BGTLevel_Spread	runtime	bgt2018, lee2016
Capex_lead	1Q lead of Capex. *Formula:* 1-quarter lead of Capex	runtime	Bates, Kahle & Stulz (2009, JF)
Capex_lead2	2Q lead of Capex. *Formula:* 2-quarter lead of Capex	runtime	Bates, Kahle & Stulz (2009, JF)
Capex_lead3	3Q lead of Capex. *Formula:* 3-quarter lead of Capex	runtime	Bates, Kahle & Stulz (2009, JF)
Capex_lead4	4Q lead of Capex. *Formula:* 4-quarter lead of Capex	runtime	Bates, Kahle & Stulz (2009, JF)
CashRatio_lead	1Q lead of CashRatio. *Formula:* 1-quarter lead of CashRatio	runtime	see base var
ChangDebtChoice_lead	1Q lead of ChangDebtChoice. *Formula:* 1-quarter lead of ChangDebtChoice	runtime	cdh2006
ChangExternalFunding_lead	1Q lead of ChangExternalFunding. *Formula:* 1-quarter lead of ChangExternalFunding	runtime	cdh2006
DebtToCapital_lead	1Q lead of DebtToCapital. *Formula:* 1-quarter lead of DebtToCapital	runtime	rz1995
DeltaILLIQ_lead1	1Q lead of DeltaILLIQ. *Formula:* 1-quarter lead of DeltaILLIQ	runtime	amihud02
DISP_lag	1Q lag of DISP. *Formula:* 1-quarter lag of DISP	runtime	wang2020

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Name	Description / Formula	Source	Reference
DISP_lead	1Q lead of DISP. *Formula:* 1-quarter lead of DISP	runtime	wang2020
DivPayerQ_lead1	1Q lead of DivPayerQ. *Formula:* 1-quarter lead of DivPayerQ	runtime	Hoberg & Prabhala (2009, RFS) – primary; Chetty & Saez (2005, QJE) – secondary (firm-quarter frequency precedent)
DSPREAD_lead1	1Q lead of DSPREAD. *Formula:* 1-quarter lead of DSPREAD	runtime	lee2016
EquityDelayCon_lead	1Q lead of EquityDelayCon. *Formula:* 1-quarter lead of EquityDelayCon	runtime	hm2015
Leverage_lead	1Q lead of Leverage. *Formula:* 1-quarter lead of Leverage	runtime	bks2009, rz1995
PayoutRatio_q_lead_qtr	1Q lead of PayoutRatio_q (qtr-aligned). *Formula:* 1-quarter lead of PayoutRatio_q (calendar-quarter aligned)	runtime	dds2004
PostCallAmihud	Panel-time post-call Amihud illiquidity level (H7b). *Formula:* Panel-time mean daily Amihud illiquidity over the post-call period (call quarter + N following quarters per H7b runner spec). Computed on panel structure with two-way clustering.	runtime	thesis (H7b panel-time post-call construction)
PostCallAmihud_lead1	1Q lead of PostCallAmihud. *Formula:* 1-quarter lead of PostCallAmihud	runtime	thesis (H7b panel-time post-call construction)
PostCallSpread	Panel-time post-call bid-ask spread level (H14b). *Formula:* Panel-time mean daily relative bid-ask spread over the post-call period (call quarter + N following quarters per H14b runner spec). Computed on panel structure with two-way clustering.	runtime	thesis (H14b panel-time post-call construction)
PostCallSpread_lead1	1Q lead of PostCallSpread. *Formula:* 1-quarter lead of PostCallSpread	runtime	thesis (H14b panel-time post-call construction)
PRisk_lag	1Q lag of PRisk. *Formula:* 1-quarter lag of PRisk	runtime	hhlt2019
PRisk_lag2	2Q lag of PRisk. *Formula:* 2-quarter lag of PRisk	runtime	hhlt2019
RDSales_lead	1Q lead of RDSales. *Formula:* 1-quarter lead of RDSales	runtime	bks2009, jjl2021
RepurchaseIntensity_lead_qtr	1Q lead of RepurchaseIntensity (qtr-aligned). *Formula:* 1-quarter lead of RepurchaseIntensity (calendar-quarter aligned)	runtime	Thesis Advisor Suggestion
UncAnsMgr_c	Mean-centered UncAnsMgr. *Formula:* UncAnsMgr minus its sample mean (mean-centered for interaction interpretability)	runtime	thesis (interaction interpretability)

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Name	Description / Formula	Source	Reference
UncAnsMgr_c_x_BelowIG	Interaction term: UncAnsMgr_c x BelowIG. *Formula:* Product: UncAnsMgr_c * BelowIG	runtime	thesis (interaction term)
UncAnsMgr_c_x_HighTSIMM	Interaction term: UncAnsMgr_c x HighTSIMM. *Formula:* Product: UncAnsMgr_c * HighTSIMM	runtime	thesis (interaction term)
UncAnsMgr_c_x_IG	Interaction term: UncAnsMgr_c x IG. *Formula:* Product: UncAnsMgr_c * IG	runtime	thesis (interaction term)
UncAnsMgr_c_x_Unrated	Interaction term: UncAnsMgr_c x Unrated. *Formula:* Product: UncAnsMgr_c * Unrated	runtime	thesis (interaction term)
UncAnsMgr_c_x_zlogTSIMM	Interaction term: UncAnsMgr_c x zlogTSIMM. *Formula:* Product: UncAnsMgr_c * zlogTSIMM	runtime	thesis (interaction term)

A.6 Stage 9

Name	Description / Formula	Source	Reference
CCCL	SEC comment letter received (binary, forward-looking) *Formula:* 1 if SEC comment letter received in next quarter, 0 otherwise	inputs/EDGAR_CommentLetters	cdm2013
DeltaILLIQ	Change in Amihud illiquidity around earnings call *Formula:* PostCallILLIQ - PreCallILLIQ over [+1,+3] vs [-3,-1] days	inputs/CRSP_DSF	amihud02
DISP	Analyst forecast dispersion (Wang 2020) *Formula:* SD(latest analyst EPS forecasts in T-31..T-1 window) / prccq_prior	inputs/tr_ibes	wang2020
DSPREAD	Change in closing bid-ask spread around earnings call *Formula:* mean(RelSpread[+1,+3]) - mean(RelSpread[-3,-1]) where RelSpread = 2*(ASK-BID)/(ASK+BID)	inputs/CRSP_DSF	lee2016
GEPU_log	Davis (2016, NBER WP 22740) Global Economic Policy Uncertainty (GEPU) index (log-transformed, current-$ GDP weights) *Formula:* log(GEPU_current) — GDP-weighted average of 16 country EPU indices, matched by calendar month of call start_date	inputs/EconomicPolicyUncertaintyIndex/Global	davis2016
GPR_log	Caldara & Iacoviello (2022, AER) headline Geopolitical Risk index (log-transformed, 10-newspaper recent version) *Formula:* log(GPR) — recent headline Geopolitical Risk index, matched by calendar month of call start_date	inputs/matteoiacoviello/GPR_Global	ci2022
SEC_Letters_fwd	SEC comment letter count (forward-looking) *Formula:* Count of SEC-originated letters (UPLOAD type) in next calendar quarter	inputs/EDGAR_CommentLetters	cdm2013 (thesis extension to count)
TotalSimilarity	Hoberg-Phillips total product similarity *Formula:* Sum of pairwise cosine similarities from 10-K product descriptions	inputs/TNIC	hp2016

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Name	Description / Formula	Source	Reference
US_EPU_log	Baker, Bloom & Davis (2016, QJE) US News-Based Economic Policy Uncertainty index (log-transformed) *Formula:* log(News_Based_Policy_Uncert_Index) matched by calendar month of call start_date	inputs/EconomicPolicyUncertaintyIndexUS	bbd2016

B Detailed Validity Tables

This appendix reports two sets of companion tables for the construct-validation and outside-world reaction analyses of §4.

B.1 Driver Regressions — Non-CEO Speech Channels

Table 7 replicates the consolidated driver matrix of §4.1 using the non-CEO manager speech channels (UncAnsNoCEO, UncPreNoCEO) in place of the CEO-partitioned regressors. A driver that moves both the CEO and non-CEO speech channels in the predicted direction is consistent with aggregate firm-call-level uncertainty content propagating to both speaker pools; a driver that loads one channel and not the other suggests channel-specific responsiveness.

Table 7: Driver Matrix — Non-CEO Speech Channels

	UncAnsNoCEO		UncPreNoCEO	
	Industry FE	Firm FE	Industry FE	Firm FE
PRisk (firm-call, Hassan et al. (2020))	**0.00014*****	**0.00012*****	**0.00014*****	**0.00011*****
	(0.00002)	(0.00002)	(0.00003)	(0.00002)
US EPU (log, Baker et al. (2016))	**0.0109***	0.0098	0.0008	0.0006
	(0.0081)	(0.0084)	(0.0079)	(0.0070)
Global EPU (log, Davis (2016))	**0.0178***	0.0166	0.0078	0.0053
	(0.0128)	(0.0132)	(0.0123)	(0.0108)
N range	68,858–73,816	68,858–73,816	73,835–76,549	73,835–76,549

Notes: Companion to Table 23. Each cell reports the coefficient and cluster-robust standard error (in parentheses) on the respective driver variable from a PanelOLS regression of the non-CEO manager speech-uncertainty regressand (UncAnsNoCEO or UncPreNoCEO) on the driver plus the standard control set. PRisk is firm-clustered; US EPU and Global EPU are two-way firm × calendar-year-quarter clustered. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$ (one-tailed positive on all drivers). Significant coefficients in **bold**.

B.2 Outside-World Reaction — Manager-Pool Coefficients

Table 8 reports the manager-pool speech-uncertainty coefficients from the same joint four-regressor specification used in §4.2. The H14c suite includes both CEO and non-CEO manager (pooled) speech-uncertainty regressors in every column; the body table reports the CEO coefficients, and this table reports the corresponding manager-pool coefficients. Dependent variable is the 26-day post-call relative bid-ask spread Spread_{25D}.

Table 8: H14c Manager-Pool Coefficients — Companion to §4.2

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
	Spread$_{25\mathrm{D}t}$						Spread$_{25\mathrm{D}t+1}$					
UncAnsMgr	-0.3223	-0.2516	**0.3616***	-0.2453	-0.2257	-0.3266	-0.5609	-0.6551	**0.5780*****	-0.4518	-0.2466	-0.3593
	(0.2454)	(0.2610)	(0.2202)	(0.2380)	(0.2121)	(0.2294)	(0.2351)	(0.2465)	(0.1819)	(0.1953)	(0.1683)	(0.1824)
UncPreMgr	-0.0123	0.1870	**0.5508*****	0.1023	-0.0070	0.0686	0.0435	0.2640	**0.9476*****	**0.2786***	0.0401	**0.2893***
	(0.1939)	(0.2529)	(0.1452)	(0.2181)	(0.1438)	(0.2112)	(0.2308)	(0.2562)	(0.1605)	(0.2124)	(0.1543)	(0.2060)
Controls	Base	Base	Ext	Ext	Ext	Ext	Base	Base	Ext	Ext	Ext	Ext
Industry FE	Yes		Yes		Yes		Yes		Yes		Yes	
Firm FE		Yes		Yes		Yes		Yes		Yes		Yes
N	63,899	63,899	44,092	44,092	44,092	44,092	63,443	63,443	44,569	44,569	44,569	44,569
R^2	0.778	0.585	0.721	0.563	0.730	0.557	0.774	0.567	0.699	0.514	0.723	0.530

Notes: Companion to body §4.2. Reports the manager-pool (UncAnsMgr, UncPreMgr) coefficients from the same 4-regressor joint specification whose CEO coefficients appear in the body. Dependent variable: 26-day post-call closing-quote relative bid-ask spread Spread$_{25\mathrm{D}}$, scaled to basis points. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$ (one-tailed positive). Significant coefficients in **bold**. Firm-clustered standard errors. Main sample 2002–2018.

C Robustness Specifications

This appendix collects auxiliary specifications and design-choice disclosures supporting the main §3 hypotheses HC and HFC.

C.1 Three-Tier Constraint Moderator: BelowIG Handling

The HFC specification of §3.3 estimates the financial-constraint moderator using a three-tier rating partition (IG, BelowIG, Unrated) but reports the IG-versus-Unrated contrast as the main-display estimand. The BelowIG category enters the model as a level shift relative to the IG reference category, with no BelowIG-specific interaction term constructed in the H1.2.ceo2 specification. This restriction reflects two design considerations. First, the FP framework on which HFC is built is binary (rated vs. unrated); the three-tier extension is our refinement, and within the rated half of the population the BelowIG cell is small and economically heterogeneous. Second, the main hypothesis localizes the precautionary cash response to the credit-constrained Unrated cell as predicted by FP, so the displayed contrast is between the credit-constrained Unrated and the rated pool (IG $\cup$ BelowIG combined as the reference). The full BelowIG level-coefficient row is reported in the body Table 14; readers wanting a fully decomposed three-tier interaction specification can re-estimate H1.2 with an explicit BelowIG $\times$ UncAnsCEO_c term added to the model.

C.2 Alternative Cash Dependent-Variable Specification (OPSW Log-Form)

The body adopts the linear cash-to-assets dependent variable (`cheq/atq`) of Bates et al. (2009) as the standard precautionary-cash form. Opler et al. (1999) also use a log-form dependent variable, defined as $\log(\text{cheq}/\text{atq})$ (or, in some specifications, $\log(\text{cheq}/(\text{atq} - \text{cheq}))$ to avoid log-of-zero issues at the lower tail). The log form down-weights the upper tail of the cash distribution relative to the linear form and tests whether the precautionary association is driven by high-cash firms.

The present draft does not estimate the OPSW log-DV robustness; it is flagged in §5 as a planned extension. The expectation, given the magnitude of the linear-form coefficients in §3.2 (UncAnsCEO sig 12/12 at $p < 0.10$), is that the qualitative result survives under the log specification, with attenuated point-estimate magnitudes due to the tail-compression. Future work will report the log-DV coefficients directly to confirm this expectation.

C.3 Fixed-Effects Ladder Robustness

The four-step fixed-effects ladder of §2.5 (Industry+Year, Firm+Year, Industry+Year-Quarter, Firm+Year-Quarter) is the within-regressor robustness frame applied throughout the body. Every column of every body table reports a different point on this ladder, so the FE-ladder robustness check is integrated rather than a separate appendix specification. The convergence pattern across the ladder—attenuation from industry to firm fixed effects, additional attenuation from year to year-quarter fixed effects—is the standard within-firm panel-OLS attenuation pattern (within-firm variation is a strict subset of cross-firm variation, so coefficients on a regressor with substantial across-firm dispersion shrink as fixed effects absorb that dispersion). The signs and significance of UncAnsCEO are robust to all four ladder steps in HC and HFC; this robustness is documented in the body and not separately tabulated here.

D Partition Decomposition: CEO vs. Non-CEO Speaker Pool

This appendix documents the empirical motivation for the CEO speaker-partition adopted throughout the body. The HC and HFC analyses of §3 use CEO-restricted speech-uncertainty regressors (UncAnsCEO, UncPreCEO) rather than the standard pooled-manager aggregate of Bushee et al. (2018). The core empirical claim that motivates this design choice is that the precautionary cash-holdings response identified in HC localizes to the CEO Q&A segment of the call: the pooled-manager regressor that mixes CEO and non-CEO speech masks this concentration.

D.1 Decomposition logic

The pooled-manager Q&A uncertainty regressor is mechanically a weighted average of the CEO-restricted regressor and a non-CEO-manager-restricted regressor, with the weights determined by the share of total Q&A words attributable to each speaker subset:

$$\text{UncAnsMgr}_c \;=\; w_c^{\text{CEO}} \cdot \text{UncAnsCEO}_c \;+\; w_c^{\text{NoCEO}} \cdot \text{UncAnsNoCEO}_c,$$

where $w_c^{\text{CEO}} + w_c^{\text{NoCEO}} = 1$ within the CEO-present sample. If the precautionary cash response loads on UncAnsCEO and UncAnsNoCEO with equal weight, the pooled-manager coefficient should approximate the equal-weighted average. If the response loads disproportionately on one channel, the pooled aggregate attenuates the channel-specific coefficient toward the population-share-weighted average and obscures the underlying mechanism.

D.2 Partition outcome taxonomy

We catalogue the partition outcomes across the suites estimated in this paper into four classes by comparing the parent (UncAnsMgr / UncPreMgr) coefficient to the partitioned (UncAnsCEO / UncPreCEO and UncAnsNoCEO / UncPreNoCEO) coefficients on the same dependent variable.

D.2.1 Class A — CEO Localization

The parent-pool signal localizes to the CEO segment under partition; the non-CEO-manager partition is null. This is the pattern observed for the cash-holdings response (HC), where UncAnsCEO is significant in 12 of 12 specifications at $p < 0.10$ while UncAnsNoCEO is materially weaker. The pooled-manager regressor would average the strong CEO signal with the weak non-CEO signal, attenuating the headline coefficient and obscuring the speaker-specific mechanism.

D.2.2 Class B — Non-CEO or Presentation-Segment Localization

The parent signal localizes to the non-CEO speaker pool or to the prepared-Presentation segment rather than to the CEO Q&A channel. This is the pattern observed in suites where forecaster or analyst reaction tracks non-CEO-manager speech rather than CEO speech, and where the CEO Q&A channel is null even when the pool-aggregated regressor is significant.

D.2.3 Class C — Signal Collapses Under Partition

The parent-pool signal disappears entirely when the regressor is partitioned by speaker, indicating that the parent coefficient was a joint-IV nesting artifact rather than a substantive economic signal. Suites in this class are not part of the body of the present paper; they are excluded from the v5.1 thesis scope on the grounds that the partition decomposition reveals the headline to be a measurement artifact.

D.2.4 Class D — Signal Preserved and Distributed

The parent-pool signal is robust and distributes approximately equally across the CEO and non-CEO partitions. This pattern is consistent with broad firm-call-level uncertainty content propagating to all manager speakers within the call. The driver regressions of §4.1 are partly in this class for PRisk (the firm-call-level political-risk index loads both CEO channels at high precision), and partly in Class B for the macro drivers (which preferentially load the Presentation channel).

D.3 HC + HFC partition verification

Table 9 summarizes the partition decomposition for the cash-holdings response (HC, suite H1) and the financial-constraint moderation (HFC, suite H1.2). Programmatic extraction from the per-suite specification files is documented in the auxiliary CSV `_partition_findings.csv` retained at the repository root for replication.

Table 9: HC + HFC Partition Decomposition Summary

Suite	Regressor partition	Contemp. sig. (at $p < 0.10$)	Lead sig. (at $p < 0.10$)	Cl
H1 (HC, parent UncAnsMgr)	Pooled manager	6 / 6	1 / 6	–
H1.r (CEO partition)	UncAnsCEO	6 / 6	6 / 6	[illegible]
H1.r (Non-CEO partition)	UncAnsNoCEO	4 / 6	0 / 6	–
H1.2 (HFC, parent UncAnsMgr×Unrated)	Pooled manager × Unrated	4 / 4	4 / 4	–
H1.2.r (CEO partition)	UncAnsCEO × Unrated	4 / 4	4 / 4	[illegible]
H1.2.r (Non-CEO partition)	UncAnsNoCEO × Unrated	0 / 4	0 / 4	–

Notes: Cells report sig-counts at $p < 0.10$ (one-tailed) for the indicated regressor across the four-step fixed-effects ladder of §2.5. “Class” assigns the partition outcome to the four-class taxonomy described in this appendix. The H1 / H1.2 results above use the auxiliary partition-variant runs (suite IDs H1.r, H1.2.r) which decompose the parent UncAnsMgr regressor into CEO and non-CEO speaker partitions on the same panel. The body of the paper estimates the CEO-restricted specifications (H1.ceo2, H1.2.ceo2) directly; this appendix table cross-validates that the cash response localizes to the CEO partition and does not load the non-CEO partition.

The pattern in Table 9 satisfies the Class A criterion for both HC and HFC: the cash response is concentrated in the CEO partition and is not detectable in the non-CEO-manager partition. The CEO-restricted regressors of §3 therefore identify the speaker locus of the precautionary signal that the standard pooled-manager construction would obscure.

Table 10: Summary Statistics

Variable	Unit	N	Mean	SD	Min	P25	Median	P75	Max
A. Independent Variables									
GEPU_log	C	88,205	4.6872	0.3672	3.8626	4.3797	4.7238	4.9666	5.5106
GPR_log	C	88,205	4.5838	0.2554	4.1614	4.4165	4.5111	4.7251	5.8825
PRisk	C	85,995	99.6187	146.5189	0.0000	17.0778	53.3729	120.1973	1192.4308
PRisk_lag	C	81,766	99.6985	146.6775	0.0000	17.1376	53.4047	120.2397	1192.4308
PRisk_lag2	C	80,627	99.9614	147.2392	0.0000	17.1702	53.4446	120.2666	1192.4308
TotalSimilarity	F	22,403	3.4277	5.6522	1.0000	1.1246	1.5865	3.0903	80.1876
US_EPU_log	C	88,205	4.7123	0.3661	3.8018	4.4805	4.6891	4.9625	5.6478
UncAnsCEO	C	71,748	0.7806	0.3925	0.0000	0.5147	0.7415	1.0018	2.5874
UncAnsMgr	C	84,729	0.8189	0.3344	0.0000	0.5891	0.7874	1.0135	2.1014
UncAnsNoCEO	C	80,387	0.8505	0.5457	0.0000	0.5042	0.7913	1.1204	3.1912
UncPreCEO	C	71,727	0.6602	0.3917	0.0000	0.3810	0.6001	0.8723	2.2432
UncPreMgr	C	86,795	0.8938	0.3508	0.0000	0.6456	0.8534	1.0952	2.1548
UncPreNoCEO	C	85,444	1.1176	0.6136	0.0000	0.7185	1.0123	1.3822	4.4188
B. Dependent Variables									
BGTAvg_Amihud	C	86,978	0.0167	0.1003	0.0000	0.0002	0.0010	0.0041	2.3333
BGTAvg_Amihud_lead1	C	82,808	0.0158	0.0980	0.0000	0.0002	0.0009	0.0038	2.3333
BGTAvg_Spread	C	87,086	0.0018	0.0027	0.0001	0.0005	0.0009	0.0018	0.0168
BGTAvg_Spread_lead1	C	82,914	0.0017	0.0025	0.0001	0.0004	0.0009	0.0017	0.0168
BGTDelta_Amihud	C	86,557	-0.0011	0.0281	-0.6913	-0.0004	-0.0000	0.0001	0.3352
BGTDelta_Amihud_lead1	C	82,475	-0.0009	0.0271	-0.6913	-0.0003	-0.0000	0.0001	0.3352
BGTDelta_Spread	C	85,189	-0.0001	0.0008	-0.0044	-0.0002	-0.0000	0.0001	0.0030
BGTDelta_Spread_lead1	C	81,145	-0.0001	0.0008	-0.0044	-0.0002	-0.0000	0.0001	0.0030
BGTLevel_Amihud	C	86,935	0.0151	0.0847	0.0000	0.0002	0.0009	0.0038	1.6839
BGTLevel_Amihud_lead1	C	82,766	0.0143	0.0831	0.0000	0.0002	0.0009	0.0035	1.6839
BGTLevel_Spread	C	87,018	0.0017	0.0026	0.0001	0.0004	0.0009	0.0017	0.0164
BGTLevel_Spread_lead1	C	82,843	0.0016	0.0024	0.0001	0.0004	0.0009	0.0016	0.0164
CCCL	C	88,205	0.0032	0.0563	0.0000	0.0000	0.0000	0.0000	1.0000
Capex	C	87,627	0.0487	0.0536	0.0000	0.0176	0.0324	0.0592	0.5203
Capex_lead	C	81,582	0.0481	0.0525	0.0000	0.0175	0.0321	0.0585	0.5203
Capex_lead2	C	73,542	0.0476	0.0513	0.0000	0.0175	0.0321	0.0581	0.5203
Capex_lead3	C	66,062	0.0472	0.0506	0.0000	0.0174	0.0319	0.0576	0.5203
Capex_lead4	C	58,987	0.0465	0.0498	0.0000	0.0172	0.0316	0.0569	0.5203
CashRatio	C	87,990	0.1708	0.1806	0.0000	0.0368	0.1047	0.2435	0.9938
CashRatio_lead	C	82,236	0.1715	0.1748	0.0000	0.0407	0.1112	0.2447	0.9938
ChangDebtChoice	C	20,448	0.7598	0.4272	0.0000	1.0000	1.0000	1.0000	1.0000
ChangDebtChoice_lead	C	18,720	0.7974	0.4019	0.0000	1.0000	1.0000	1.0000	1.0000
ChangExternalFunding	C	87,532	0.2506	0.4334	0.0000	0.0000	0.0000	1.0000	1.0000
ChangExternalFunding_lead	C	81,563	0.2446	0.4299	0.0000	0.0000	0.0000	0.0000	1.0000
DISP	C	37,446	0.0017	0.0028	0.0000	0.0003	0.0007	0.0018	0.0179
DISP_lead	C	37,015	0.0017	0.0027	0.0000	0.0003	0.0007	0.0018	0.0179
DSPREAD	C	87,119	-0.0002	0.0017	-0.0081	-0.0004	-0.0000	0.0001	0.0064
DSPREAD_lead1	C	82,964	-0.0002	0.0016	-0.0081	-0.0004	-0.0000	0.0001	0.0064
DebtToCapital	C	87,760	0.3450	0.3541	0.0000	0.0689	0.3064	0.4914	4.4691
DebtToCapital_lead	C	82,024	0.3495	0.3536	0.0000	0.0773	0.3110	0.4958	4.4691
DeltaILLIQ	C	86,906	-0.0013	0.0249	-0.5050	-0.0006	-0.0000	0.0001	0.3003
DeltaILLIQ_lead1	C	82,735	-0.0012	0.0243	-0.5050	-0.0005	-0.0000	0.0001	0.3003
DivPayerQ	C	88,134	0.4235	0.4941	0.0000	0.0000	0.0000	1.0000	1.0000
DivPayerQ_lead1	C	83,755	0.4290	0.4949	0.0000	0.0000	0.0000	1.0000	1.0000
EquityDelayCon_lead	F	13,212	-0.0270	0.0772	-0.2577	-0.0814	-0.0293	0.0233	0.3006
Leverage	C	87,994	0.2330	0.2240	0.0000	0.0470	0.2065	0.3455	3.9453
Leverage_lead	C	82,242	0.2352	0.2248	0.0000	0.0530	0.2090	0.3480	3.9453
PayoutRatio_q	C	70,695	0.2420	0.6534	0.0000	0.0000	0.0000	0.2794	9.2194
PayoutRatio_q_lead_qtr	C	68,167	0.2437	0.6562	0.0000	0.0000	0.0000	0.2816	9.2194
PostCallAmihud	C	86,906	0.0118	0.0680	0.0000	0.0002	0.0007	0.0031	1.9001
PostCallAmihud_lead1	C	82,735	0.0112	0.0669	0.0000	0.0002	0.0007	0.0028	1.9001
PostCallSpread	C	87,119	0.0017	0.0028	-0.0075	0.0004	0.0008	0.0017	0.0250
PostCallSpread_lead1	C	82,964	0.0016	0.0027	-0.0075	0.0004	0.0008	0.0016	0.0250
RDSales	C	87,647	0.1047	0.9342	0.0000	0.0000	0.0048	0.0636	39.7688
RDSales_lead	C	81,607	0.0990	0.9084	0.0000	0.0000	0.0050	0.0638	39.7688
RepurchaseIntensity	C	83,734	0.0073	0.0148	0.0000	0.0000	0.0000	0.0074	0.0919
RepurchaseIntensity_lead_qtr	C	80,289	0.0075	0.0149	0.0000	0.0000	0.0000	0.0078	0.0919
SEC_Letters_fwd	C	88,205	0.1264	0.4873	0.0000	0.0000	0.0000	0.0000	12.0000
C. Firm Controls									
AbsSurpDec	C	61,544	2.9300	1.4255	0.0000	2.0000	3.0000	4.0000	5.0000

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Variable	Unit	N	Mean	SD	Min	P25	Median	P75	Max
CashFlowAt	C	87,670	0.1019	0.1147	-3.0324	0.0608	0.1032	0.1526	0.4874
DISP_lag	C	36,411	0.0016	0.0027	0.0000	0.0003	0.0007	0.0017	0.0179
DailyVola	C	82,599	36.7556	21.2437	9.4388	22.9709	31.2762	43.8148	211.9160
DivDummy	C	88,134	0.4755	0.4994	0.0000	0.0000	0.0000	1.0000	1.0000
EarnVol	C	87,440	0.0701	0.1907	0.0006	0.0163	0.0315	0.0695	19.5101
FirmMat	C	87,571	-0.0432	2.1826	-317.5714	-0.0078	0.2191	0.4195	0.9252
Loss	C	88,129	0.1953	0.3964	0.0000	0.0000	0.0000	0.0000	1.0000
NegCall	C	86,858	0.9210	0.3024	0.0000	0.7038	0.8798	1.0943	2.3901
ROA	C	87,552	0.0386	0.1330	-3.9826	0.0146	0.0525	0.0925	0.5490
SalesGrowth	C	87,506	0.1173	0.3849	-1.0000	-0.0043	0.0745	0.1762	10.3051
StockPrice	C	87,188	38.6939	51.7195	0.0345	14.8975	28.2400	48.1700	2080.0200
SurpDec	C	61,544	1.2085	3.0260	-5.0000	-1.0000	2.0000	4.0000	5.0000
TobinsQ	C	87,363	2.1363	1.5197	0.3847	1.2681	1.6791	2.4362	35.6144
Turnover	C	87,188	0.0273	0.0420	0.0000	0.0083	0.0160	0.0310	3.7726
UncQue	C	81,031	1.4398	0.4831	0.0000	1.1216	1.4108	1.7268	3.5945
lnAssets	C	87,994	7.4710	1.6607	-0.5692	6.3008	7.3763	8.5436	12.4592
sCFO	C	83,662	0.0619	0.2233	0.0014	0.0236	0.0395	0.0664	32.4035

Notes: Summary statistics for variables used in the hypothesis tests. Sample period: 2002–2018. Anchor sample (H1, Main): 88,205 earnings-call observations across 1,884 firms. Unit column: C = call-level, F = firm-year. N varies by row because each variable is summarized on its anchor panel's Main sample (complete cases per variable); per-suite samples differ due to lags, control availability, and merges.

Table 11: H1 CEO 2-IV: CEO Q&A + Presentation Uncertainty

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
	CashRatio						CashRatio_lead					
UncAnsCEO	**0.0027*****	**0.0020*****	**0.0030*****	**0.0018****	**0.0031*****	**0.0019****	**0.0023***	**0.0028****	**0.0025***	**0.0025****	**0.0024***	**0.0024****
	(0.0010)	(0.0008)	(0.0010)	(0.0008)	(0.0010)	(0.0008)	(0.0016)	(0.0012)	(0.0016)	(0.0012)	(0.0016)	(0.0012)
UncPreCEO	0.0001	0.0002	0.0002	0.0000	0.0002	0.0002	0.0006	0.0016	0.0006	0.0013	0.0006	0.0012
	(0.0010)	(0.0010)	(0.0010)	(0.0010)	(0.0010)	(0.0010)	(0.0019)	(0.0016)	(0.0019)	(0.0015)	(0.0019)	(0.0015)
Leverage	**-0.0196*****	**-0.0352*****	**-0.0143*****	**-0.0247*****	**-0.0147*****	**-0.0250*****	**-0.0366*****	**-0.0242*****	**-0.0305*****	**-0.0161***	**-0.0305*****	**-0.0155***
	(0.0039)	(0.0061)	(0.0031)	(0.0062)	(0.0032)	(0.0062)	(0.0099)	(0.0090)	(0.0085)	(0.0091)	(0.0085)	(0.0091)
lnAssets	**-0.0016*****	**-0.0165*****	**-0.0019*****	**-0.0173*****	**-0.0017*****	**-0.0169*****	**-0.0028*****	**-0.0290*****	**-0.0027*****	**-0.0292*****	**-0.0027*****	**-0.0293*****
	(0.0004)	(0.0018)	(0.0005)	(0.0018)	(0.0005)	(0.0018)	(0.0008)	(0.0030)	(0.0008)	(0.0031)	(0.0008)	(0.0031)
TobinsQ	**0.0057*****	**0.0051*****	**0.0048*****	**0.0041*****	**0.0047*****	**0.0038*****	**0.0067*****	0.0019	**0.0054*****	0.0012	**0.0054*****	0.0013
	(0.0006)	(0.0011)	(0.0006)	(0.0013)	(0.0006)	(0.0013)	(0.0012)	(0.0014)	(0.0010)	(0.0015)	(0.0010)	(0.0015)
ROA	**-0.0141****	0.0122	**-0.0624*****	**-0.0296*****	**-0.0593*****	**-0.0277*****	**-0.0619*****	**-0.0272****	**-0.1330*****	**-0.0716*****	**-0.1292*****	**-0.0696*****
	(0.0062)	(0.0082)	(0.0078)	(0.0094)	(0.0078)	(0.0095)	(0.0139)	(0.0135)	(0.0134)	(0.0152)	(0.0134)	(0.0153)
Capex	**-0.1068*****	**-0.2294*****	**-0.1431*****	**-0.2399*****	**-0.1400*****	**-0.2315*****	**-0.1521*****	**-0.2140*****	**-0.2209*****	**-0.2308*****	**-0.2190*****	**-0.2330*****
	(0.0112)	(0.0225)	(0.0131)	(0.0233)	(0.0130)	(0.0232)	(0.0178)	(0.0339)	(0.0200)	(0.0336)	(0.0200)	(0.0338)
DivDummy	**-0.0044*****	-0.0018	**-0.0050*****	-0.0026	**-0.0045*****	-0.0022	**-0.0067*****	-0.0052	**-0.0063*****	-0.0053	**-0.0062*****	-0.0055
	(0.0011)	(0.0018)	(0.0011)	(0.0018)	(0.0011)	(0.0018)	(0.0021)	(0.0036)	(0.0020)	(0.0035)	(0.0020)	(0.0036)
sCFO	**0.0120****	0.0010	**0.0137****	0.0018	**0.0133****	0.0017	**0.0159****	0.0047	**0.0173****	0.0052	**0.0170****	0.0051
	(0.0054)	(0.0022)	(0.0068)	(0.0026)	(0.0066)	(0.0025)	(0.0072)	(0.0033)	(0.0085)	(0.0034)	(0.0083)	(0.0034)
Lagged_DV	**0.8508*****	**0.6289*****	**0.8555*****	**0.6395*****	**0.8560*****	**0.6407*****	**0.7114*****	**0.2198*****	**0.7170*****	**0.2314*****	**0.7174*****	**0.2318*****
	(0.0065)	(0.0102)	(0.0067)	(0.0101)	(0.0067)	(0.0101)	(0.0127)	(0.0154)	(0.0123)	(0.0156)	(0.0123)	(0.0156)
SalesGrowth			**-0.0123*****	**-0.0175*****	**-0.0115*****	**-0.0168*****			**-0.0130*****	**-0.0163*****	**-0.0121****	**-0.0159*****
			(0.0033)	(0.0034)	(0.0032)	(0.0034)			(0.0048)	(0.0037)	(0.0048)	(0.0037)
RDSales			**0.0045***	-0.0003	**0.0045***	-0.0002			**0.0066***	-0.0048	**0.0066***	-0.0047
			(0.0023)	(0.0007)	(0.0023)	(0.0007)			(0.0035)	(0.0030)	(0.0035)	(0.0030)
CashFlowAt			**0.1034*****	**0.1273*****	**0.1029*****	**0.1266*****			**0.1635*****	**0.1445*****	**0.1601*****	**0.1418*****
			(0.0121)	(0.0160)	(0.0122)	(0.0161)			(0.0178)	(0.0190)	(0.0178)	(0.0191)
DailyVola			0.0000	**-0.0001*****	**0.0001*****	-0.0000			**0.0002*****	**0.0001*****	**0.0002*****	**0.0001***
			(0.0000)	(0.0000)	(0.0000)	(0.0000)			(0.0000)	(0.0000)	(0.0001)	(0.0000)
Extended Controls			Yes	Yes	Yes	Yes			Yes	Yes	Yes	Yes
Industry FE	Yes		Yes		Yes		Yes		Yes		Yes	
Firm FE		Yes		Yes		Yes		Yes		Yes		Yes
Year FE	Yes	Yes	Yes	Yes			Yes	Yes	Yes	Yes		
Year-Quarter FE					Yes	Yes					Yes	Yes
N	65,148	65,148	62,523	62,523	62,523	62,523	60,638	60,638	59,459	59,459	59,459	59,459
R^2	0.819	0.452	0.822	0.457	0.823	0.458	0.636	0.093	0.645	0.106	0.646	0.106
Adj. R^2	0.819	0.437	0.822	0.442	0.823	0.442	0.636	0.067	0.645	0.080	0.645	0.079

Notes: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$ (one-tailed for IVs, $\beta > 0$; two-tailed for controls). Significant coefficients in **bold**. Standard errors (in parentheses) clustered at firm level. Main sample (excludes financial and utility firms). R^2 includes absorbed fixed effects (not within-R^2).

Table 12: H1 CEO 3-IV Decomp: ClarityCEO + UncResCEO (DWZ Full) + UncPreCEO

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
	CashRatio						CashRatio_lead					
CEO Clarity (DWZ)	**-0.0046***	**-0.0071***	**-0.0058****	-0.0054	**-0.0056****	-0.0052	-0.0052	**-0.0170****	**-0.0068***	**-0.0162****	**-0.0069***	**-0.0161****
	(0.0033)	(0.0051)	(0.0033)	(0.0048)	(0.0033)	(0.0047)	(0.0052)	(0.0095)	(0.0051)	(0.0090)	(0.0051)	(0.0090)
CEO Residual Unc. (DWZ)	**0.0021****	**0.0019****	**0.0020****	**0.0016***	**0.0021****	**0.0016***	**0.0028****	**0.0023****	**0.0026****	**0.0020***	**0.0025****	**0.0019***
	(0.0010)	(0.0010)	(0.0010)	(0.0010)	(0.0010)	(0.0010)	(0.0014)	(0.0013)	(0.0014)	(0.0013)	(0.0014)	(0.0013)
CEO Pres Uncertainty	0.0002	0.0009	0.0003	0.0006	0.0004	0.0007	-0.0003	0.0008	-0.0005	0.0004	-0.0007	0.0002
	(0.0011)	(0.0011)	(0.0011)	(0.0011)	(0.0011)	(0.0011)	(0.0021)	(0.0017)	(0.0021)	(0.0017)	(0.0021)	(0.0017)
Leverage	**-0.0177*****	**-0.0355*****	**-0.0130*****	**-0.0218*****	**-0.0132*****	**-0.0216*****	**-0.0327*****	**-0.0279****	**-0.0274*****	-0.0159	**-0.0275*****	-0.0156
	(0.0048)	(0.0084)	(0.0036)	(0.0082)	(0.0037)	(0.0082)	(0.0120)	(0.0121)	(0.0104)	(0.0118)	(0.0105)	(0.0118)
lnAssets	**-0.0015*****	**-0.0177*****	**-0.0017*****	**-0.0175*****	**-0.0014*****	**-0.0170*****	**-0.0024****	**-0.0268*****	**-0.0020****	**-0.0265*****	**-0.0020****	**-0.0265*****
	(0.0006)	(0.0026)	(0.0005)	(0.0026)	(0.0005)	(0.0026)	(0.0010)	(0.0042)	(0.0009)	(0.0042)	(0.0010)	(0.0042)
TobinsQ	**0.0058*****	**0.0048*****	**0.0042*****	**0.0037*****	**0.0040*****	**0.0034*****	**0.0085*****	0.0025	**0.0056*****	0.0009	**0.0056*****	0.0010
	(0.0006)	(0.0009)	(0.0007)	(0.0010)	(0.0007)	(0.0010)	(0.0012)	(0.0015)	(0.0012)	(0.0015)	(0.0012)	(0.0016)
ROA	-0.0144	**0.0214****	**-0.0679*****	-0.0195	**-0.0630*****	-0.0171	**-0.0762*****	**-0.0352***	**-0.1427*****	**-0.0790*****	**-0.1378*****	**-0.0769*****
	(0.0093)	(0.0105)	(0.0113)	(0.0122)	(0.0114)	(0.0122)	(0.0161)	(0.0185)	(0.0173)	(0.0194)	(0.0174)	(0.0196)
Capex	**-0.1117*****	**-0.2995*****	**-0.1596*****	**-0.3074*****	**-0.1564*****	**-0.2993*****	**-0.1613*****	**-0.3281*****	**-0.2433*****	**-0.3415*****	**-0.2411*****	**-0.3430*****
	(0.0128)	(0.0234)	(0.0149)	(0.0236)	(0.0148)	(0.0234)	(0.0184)	(0.0344)	(0.0216)	(0.0345)	(0.0216)	(0.0346)
DivDummy	**-0.0041*****	-0.0028	**-0.0054*****	**-0.0041***	**-0.0049*****	-0.0036	**-0.0051****	-0.0068	**-0.0056****	**-0.0076***	**-0.0054****	**-0.0077***
	(0.0013)	(0.0023)	(0.0013)	(0.0022)	(0.0013)	(0.0022)	(0.0023)	(0.0045)	(0.0023)	(0.0044)	(0.0023)	(0.0044)
sCFO	**0.0066*****	-0.0022	**0.0079*****	-0.0008	**0.0074*****	-0.0009	**0.0137****	0.0027	**0.0137****	0.0035	**0.0134****	0.0034
	(0.0020)	(0.0014)	(0.0030)	(0.0014)	(0.0028)	(0.0014)	(0.0057)	(0.0025)	(0.0062)	(0.0027)	(0.0061)	(0.0027)
Lagged_DV	**0.8558*****	**0.6146*****	**0.8612*****	**0.6322*****	**0.8617*****	**0.6337*****	**0.7214*****	**0.2209*****	**0.7227*****	**0.2377*****	**0.7231*****	**0.2385*****
	(0.0076)	(0.0133)	(0.0075)	(0.0130)	(0.0075)	(0.0130)	(0.0138)	(0.0201)	(0.0133)	(0.0201)	(0.0133)	(0.0201)
SalesGrowth			**-0.0178*****	**-0.0274*****	**-0.0169*****	**-0.0266*****			**-0.0156****	**-0.0171*****	**-0.0144****	**-0.0163*****
			(0.0045)	(0.0052)	(0.0044)	(0.0052)			(0.0061)	(0.0050)	(0.0060)	(0.0050)
RDSales			**0.0062*****	**-0.0032*****	**0.0062*****	**-0.0031****			**0.0137*****	-0.0014	**0.0138*****	-0.0012
			(0.0017)	(0.0012)	(0.0017)	(0.0012)			(0.0024)	(0.0019)	(0.0024)	(0.0019)
CashFlowAt			**0.1292*****	**0.1589*****	**0.1278*****	**0.1576*****			**0.1982*****	**0.1687*****	**0.1944*****	**0.1660*****
			(0.0162)	(0.0173)	(0.0162)	(0.0173)			(0.0236)	(0.0249)	(0.0237)	(0.0250)
DailyVola			0.0000	**-0.0001*****	**0.0001*****	**-0.0001***			**0.0003*****	**0.0001*****	**0.0003*****	0.0001
			(0.0000)	(0.0000)	(0.0000)	(0.0000)			(0.0000)	(0.0000)	(0.0001)	(0.0001)
Extended Controls			Yes	Yes	Yes	Yes			Yes	Yes	Yes	Yes
Industry FE	Yes		Yes		Yes		Yes		Yes		Yes	
Firm FE		Yes		Yes		Yes		Yes		Yes		Yes
Year FE	Yes	Yes	Yes	Yes			Yes	Yes	Yes	Yes		
Year-Quarter FE					Yes	Yes					Yes	Yes
N	43,333	43,333	43,316	43,316	43,316	43,316	41,122	41,122	41,108	41,108	41,108	41,108
R^2	0.825	0.431	0.829	0.449	0.830	0.449	0.659	0.092	0.667	0.108	0.668	0.108
Adj. R^2	0.825	0.412	0.829	0.430	0.830	0.430	0.658	0.062	0.667	0.078	0.667	0.077

Notes: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$ (one-tailed for IVs, $\beta > 0$; two-tailed for controls). Significant coefficients in **bold**. Standard errors (in parentheses) clustered at firm level. Main sample (excludes financial and utility firms). R^2 includes absorbed fixed effects (not within-R^2).

Table 13: H1 CEO 3-IV Decomp: ClarityCEO + UncResCEO (DWZ Qtr-Exp, no-look-ahead) + UncPreCEO

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
	CashRatio						CashRatio_lead					
CEO Clarity (DWZ Qtr-Exp)	**-0.0083*****	**-0.0148*****	**-0.0093*****	**-0.0129*****	**-0.0092*****	**-0.0126*****	-0.0024	**-0.0215****	-0.0041	**-0.0210****	-0.0042	**-0.0210****
	(0.0033)	(0.0047)	(0.0033)	(0.0045)	(0.0033)	(0.0045)	(0.0053)	(0.0097)	(0.0053)	(0.0094)	(0.0053)	(0.0095)
CEO Residual Unc. (DWZ Qtr-Exp)	**0.0018***	**0.0020****	**0.0016***	**0.0018***	**0.0016***	**0.0017***	**0.0046*****	**0.0036****	**0.0044****	**0.0033****	**0.0044****	**0.0033****
	(0.0012)	(0.0012)	(0.0012)	(0.0012)	(0.0012)	(0.0012)	(0.0019)	(0.0017)	(0.0019)	(0.0017)	(0.0019)	(0.0017)
CEO Pres Uncertainty	0.0001	0.0006	-0.0001	0.0000	-0.0001	0.0001	0.0002	0.0016	-0.0003	0.0011	-0.0004	0.0011
	(0.0013)	(0.0013)	(0.0013)	(0.0013)	(0.0013)	(0.0013)	(0.0024)	(0.0021)	(0.0024)	(0.0021)	(0.0024)	(0.0021)
Leverage	**-0.0131*****	**-0.0324*****	**-0.0089*****	**-0.0199****	**-0.0087*****	**-0.0190****	**-0.0293****	**-0.0295****	**-0.0241****	-0.0181	**-0.0244****	-0.0179
	(0.0040)	(0.0086)	(0.0031)	(0.0085)	(0.0031)	(0.0085)	(0.0122)	(0.0139)	(0.0108)	(0.0136)	(0.0109)	(0.0136)
lnAssets	-0.0009	**-0.0169*****	**-0.0011***	**-0.0169*****	**-0.0011***	**-0.0167*****	**-0.0018***	**-0.0285*****	-0.0016	**-0.0286*****	-0.0015	**-0.0287*****
	(0.0006)	(0.0028)	(0.0006)	(0.0027)	(0.0006)	(0.0027)	(0.0011)	(0.0049)	(0.0010)	(0.0049)	(0.0011)	(0.0050)
TobinsQ	**0.0052*****	**0.0039*****	**0.0034*****	**0.0028****	**0.0033*****	**0.0025****	**0.0079*****	0.0011	**0.0046*****	-0.0009	**0.0046*****	-0.0009
	(0.0007)	(0.0011)	(0.0009)	(0.0012)	(0.0009)	(0.0012)	(0.0012)	(0.0017)	(0.0014)	(0.0017)	(0.0014)	(0.0018)
ROA	-0.0107	**0.0230***	**-0.0510*****	-0.0135	**-0.0498*****	-0.0129	**-0.0621*****	-0.0188	**-0.1280*****	**-0.0721*****	**-0.1270*****	**-0.0731*****
	(0.0109)	(0.0124)	(0.0131)	(0.0144)	(0.0132)	(0.0144)	(0.0192)	(0.0227)	(0.0202)	(0.0237)	(0.0204)	(0.0240)
Capex	**-0.0967*****	**-0.2472*****	**-0.1415*****	**-0.2628*****	**-0.1414*****	**-0.2605*****	**-0.1613*****	**-0.2816*****	**-0.2420*****	**-0.3002*****	**-0.2423*****	**-0.2996*****
	(0.0150)	(0.0253)	(0.0169)	(0.0258)	(0.0168)	(0.0257)	(0.0214)	(0.0372)	(0.0249)	(0.0375)	(0.0249)	(0.0375)
DivDummy	**-0.0037****	-0.0021	**-0.0051*****	-0.0034	**-0.0050*****	-0.0033	**-0.0042***	-0.0080	**-0.0048****	**-0.0090***	**-0.0047***	**-0.0091***
	(0.0015)	(0.0026)	(0.0015)	(0.0025)	(0.0015)	(0.0025)	(0.0025)	(0.0054)	(0.0025)	(0.0053)	(0.0025)	(0.0053)
sCFO	**0.0090*****	-0.0008	**0.0100*****	0.0004	**0.0099*****	0.0004	**0.0147***	0.0031	**0.0143***	0.0038	**0.0142***	0.0037
	(0.0024)	(0.0017)	(0.0031)	(0.0014)	(0.0030)	(0.0015)	(0.0081)	(0.0032)	(0.0082)	(0.0032)	(0.0081)	(0.0032)
Lagged_DV	**0.8718*****	**0.6433*****	**0.8734*****	**0.6576*****	**0.8738*****	**0.6579*****	**0.7424*****	**0.2158*****	**0.7404*****	**0.2320*****	**0.7402*****	**0.2317*****
	(0.0091)	(0.0159)	(0.0091)	(0.0156)	(0.0091)	(0.0156)	(0.0150)	(0.0242)	(0.0148)	(0.0242)	(0.0148)	(0.0242)
SalesGrowth			**-0.0165*****	**-0.0257*****	**-0.0165*****	**-0.0256*****			-0.0106	**-0.0137****	-0.0107	**-0.0136****
			(0.0054)	(0.0068)	(0.0054)	(0.0068)			(0.0070)	(0.0064)	(0.0070)	(0.0064)
RDSales			**0.0091*****	-0.0004	**0.0091*****	-0.0003			**0.0160*****	0.0005	**0.0159*****	0.0005
			(0.0025)	(0.0017)	(0.0025)	(0.0017)			(0.0038)	(0.0021)	(0.0038)	(0.0021)
CashFlowAt			**0.1182*****	**0.1536*****	**0.1189*****	**0.1544*****			**0.1977*****	**0.1883*****	**0.1974*****	**0.1880*****
			(0.0159)	(0.0175)	(0.0159)	(0.0175)			(0.0275)	(0.0300)	(0.0275)	(0.0300)
DailyVola			0.0000	**-0.0001****	0.0000	**-0.0001****			**0.0002*****	0.0000	**0.0003*****	0.0000
			(0.0000)	(0.0000)	(0.0000)	(0.0000)			(0.0001)	(0.0000)	(0.0001)	(0.0001)
Extended Controls			Yes	Yes	Yes	Yes			Yes	Yes	Yes	Yes
Industry FE	Yes		Yes		Yes		Yes		Yes		Yes	
Firm FE		Yes		Yes		Yes		Yes		Yes		Yes
Year FE	Yes	Yes	Yes	Yes			Yes	Yes	Yes	Yes		
Year-Quarter FE					Yes	Yes					Yes	Yes
N	24,868	24,868	24,855	24,855	24,855	24,855	23,127	23,127	23,120	23,120	23,120	23,120
R^2	0.839	0.457	0.842	0.473	0.842	0.473	0.675	0.087	0.683	0.105	0.683	0.105
Adj. R^2	0.839	0.430	0.841	0.446	0.842	0.446	0.674	0.041	0.682	0.059	0.682	0.058

Notes: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$ (one-tailed for IVs, $\beta > 0$; two-tailed for controls). Significant coefficients in **bold**. Standard errors (in parentheses) clustered at firm level. Main sample (excludes financial and utility firms). R^2 includes absorbed fixed effects (not within-R^2).

Table 14: H1.2 CEO 2-IV: Financial Constraint–Moderated CEO Speech Uncertainty and Cash Holdings

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	CashRatio				CashRatio_lead			
UncAnsCEO_c	**0.0029*****	**0.0017****	**0.0030*****	**0.0018****	**0.0022***	**0.0024****	**0.0021***	**0.0023****
	(0.0010)	(0.0009)	(0.0010)	(0.0009)	(0.0016)	(0.0013)	(0.0016)	(0.0013)
UncPreCEO_c	0.0004	0.0001	0.0004	0.0002	0.0008	0.0012	0.0007	0.0011
	(0.0011)	(0.0011)	(0.0011)	(0.0011)	(0.0019)	(0.0016)	(0.0019)	(0.0016)
Unrated	**0.0066*****	-0.0012	-0.0013	-0.0012	**0.0103*****	-0.0043	-0.0053	-0.0043
	(0.0012)	(0.0034)	(0.0017)	(0.0034)	(0.0022)	(0.0061)	(0.0031)	(0.0061)
UncAnsCEO_c_x_Unrated	**0.0030***	0.0010	**0.0032****	0.0011	**0.0068****	**0.0040***	**0.0069****	**0.0041****
	(0.0019)	(0.0017)	(0.0019)	(0.0017)	(0.0031)	(0.0025)	(0.0031)	(0.0025)
UncPreCEO_c_x_Unrated	**0.0028***	0.0010	**0.0027***	0.0010	0.0027	-0.0006	0.0026	-0.0007
	(0.0020)	(0.0021)	(0.0019)	(0.0021)	(0.0035)	(0.0032)	(0.0035)	(0.0032)
Leverage	**-0.0140*****	**-0.0229*****	**-0.0150*****	**-0.0232*****	**-0.0261*****	-0.0117	**-0.0281*****	-0.0112
	(0.0033)	(0.0069)	(0.0035)	(0.0069)	(0.0089)	(0.0097)	(0.0095)	(0.0097)
lnAssets	**0.0005****	**-0.0169*****	**-0.0021*****	**-0.0165*****	**0.0022*****	**-0.0302*****	**-0.0035*****	**-0.0303*****
	(0.0002)	(0.0020)	(0.0006)	(0.0020)	(0.0004)	(0.0033)	(0.0010)	(0.0033)
TobinsQ	**0.0054*****	**0.0049*****	**0.0048*****	**0.0045*****	**0.0057*****	0.0008	**0.0047*****	0.0010
	(0.0007)	(0.0014)	(0.0007)	(0.0014)	(0.0010)	(0.0015)	(0.0010)	(0.0015)
ROA	**-0.0620*****	**-0.0275*****	**-0.0575*****	**-0.0256*****	**-0.1351*****	**-0.0736*****	**-0.1282*****	**-0.0716*****
	(0.0081)	(0.0098)	(0.0080)	(0.0098)	(0.0138)	(0.0158)	(0.0137)	(0.0159)
Capex	**-0.1360*****	**-0.2392*****	**-0.1377*****	**-0.2305*****	**-0.2057*****	**-0.2275*****	**-0.2143*****	**-0.2298*****
	(0.0131)	(0.0249)	(0.0134)	(0.0248)	(0.0202)	(0.0354)	(0.0210)	(0.0357)
DivDummy	**-0.0049*****	**-0.0034***	**-0.0052*****	-0.0030	**-0.0057*****	-0.0049	**-0.0072*****	-0.0051
	(0.0012)	(0.0019)	(0.0012)	(0.0019)	(0.0020)	(0.0035)	(0.0021)	(0.0036)
sCFO	**0.0116****	0.0015	**0.0106****	0.0015	**0.0174****	0.0061	**0.0158****	0.0060
	(0.0053)	(0.0024)	(0.0047)	(0.0024)	(0.0087)	(0.0038)	(0.0076)	(0.0038)
SalesGrowth	**-0.0121*****	**-0.0167*****	**-0.0109*****	**-0.0159*****	**-0.0122****	**-0.0145*****	**-0.0108****	**-0.0140*****
	(0.0036)	(0.0035)	(0.0035)	(0.0035)	(0.0051)	(0.0036)	(0.0050)	(0.0037)
RDSales	0.0041	0.0013	0.0041	0.0015	**0.0062***	-0.0052	0.0062	-0.0051
	(0.0025)	(0.0016)	(0.0025)	(0.0016)	(0.0038)	(0.0042)	(0.0038)	(0.0042)
CashFlowAt	**0.1032*****	**0.1294*****	**0.1002*****	**0.1287*****	**0.1697*****	**0.1462*****	**0.1606*****	**0.1434*****
	(0.0128)	(0.0175)	(0.0128)	(0.0176)	(0.0183)	(0.0197)	(0.0182)	(0.0197)
DailyVola	**0.0000***	**-0.0001*****	**0.0001***	-0.0000	**0.0004*****	**0.0001*****	**0.0002*****	0.0001
	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0001)	(0.0000)
Lagged_DV	**0.8580*****	**0.6285*****	**0.8542*****	**0.6298*****	**0.7272*****	**0.2179*****	**0.7186*****	**0.2185*****
	(0.0069)	(0.0104)	(0.0069)	(0.0104)	(0.0124)	(0.0150)	(0.0126)	(0.0151)
Industry FE	Yes		Yes		Yes		Yes	
Firm FE		Yes		Yes		Yes		Yes
Year FE	Yes	Yes			Yes	Yes		
Year-Quarter FE			Yes	Yes			Yes	Yes
N	56,795	56,795	56,795	56,795	55,692	55,692	55,692	55,692
R^2	0.820	0.442	0.821	0.442	0.641	0.098	0.644	0.098
Adj. R^2	0.819	0.425	0.821	0.425	0.641	0.071	0.643	0.070

Notes: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$ (one-tailed for IVs, $\beta > 0$; two-tailed for controls). Significant coefficients in **bold**. Standard errors (in parentheses) clustered at firm level. Main sample (excludes financial and utility firms). Fiscal years 2002-2016. R^2 includes absorbed fixed effects (not within-R^2).

Table 15: H1.2 CEO 3-IV Decomp HFC: ClarityCEO + UncResCEO (DWZ Full) + UncPreCEO × Unrated Constraint

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	CashRatio				CashRatio_lead			
CEO Clarity (DWZ, c)	**-0.0062****	-0.0043	**-0.0059****	-0.0042	-0.0052	**-0.0157***	-0.0052	**-0.0155***
	(0.0034)	(0.0052)	(0.0034)	(0.0052)	(0.0053)	(0.0096)	(0.0053)	(0.0096)
CEO Residual Unc. (DWZ, c)	**0.0020****	**0.0018****	**0.0022****	**0.0019****	**0.0027****	**0.0022***	**0.0026****	**0.0021***
	(0.0011)	(0.0010)	(0.0011)	(0.0010)	(0.0015)	(0.0013)	(0.0015)	(0.0013)
CEO Pres Unc. (c)	0.0002	0.0005	0.0003	0.0007	-0.0007	-0.0001	-0.0008	-0.0002
	(0.0012)	(0.0012)	(0.0012)	(0.0012)	(0.0022)	(0.0018)	(0.0022)	(0.0018)
Unrated	**0.0038*****	-0.0027	-0.0010	-0.0026	**0.0065*****	**-0.0093***	-0.0014	**-0.0095***
	(0.0012)	(0.0035)	(0.0016)	(0.0035)	(0.0022)	(0.0054)	(0.0029)	(0.0054)
UncRes × Unrated	0.0011	-0.0001	0.0012	-0.0000	**0.0057****	0.0033	**0.0057****	0.0033
	(0.0021)	(0.0020)	(0.0021)	(0.0020)	(0.0029)	(0.0026)	(0.0029)	(0.0026)
UncPre × Unrated	0.0024	0.0013	**0.0029***	0.0014	0.0005	-0.0026	0.0010	-0.0027
	(0.0022)	(0.0025)	(0.0023)	(0.0025)	(0.0040)	(0.0038)	(0.0040)	(0.0038)
Leverage	**-0.0117*****	**-0.0216****	**-0.0141*****	**-0.0214****	**-0.0237****	-0.0107	**-0.0274****	-0.0104
	(0.0033)	(0.0098)	(0.0041)	(0.0099)	(0.0103)	(0.0131)	(0.0117)	(0.0131)
lnAssets	**0.0005***	**-0.0159*****	**-0.0019*****	**-0.0155*****	**0.0018*****	**-0.0273*****	**-0.0025****	**-0.0275*****
	(0.0003)	(0.0027)	(0.0007)	(0.0027)	(0.0005)	(0.0044)	(0.0011)	(0.0044)
TobinsQ	**0.0047*****	**0.0045*****	**0.0041*****	**0.0040*****	**0.0059*****	0.0009	**0.0052*****	0.0010
	(0.0008)	(0.0011)	(0.0008)	(0.0011)	(0.0013)	(0.0017)	(0.0013)	(0.0017)
ROA	**-0.0715*****	-0.0157	**-0.0651*****	-0.0133	**-0.1452*****	**-0.0819*****	**-0.1378*****	**-0.0800*****
	(0.0122)	(0.0122)	(0.0122)	(0.0122)	(0.0182)	(0.0200)	(0.0183)	(0.0202)
Capex	**-0.1610*****	**-0.3087*****	**-0.1647*****	**-0.3005*****	**-0.2338*****	**-0.3399*****	**-0.2426*****	**-0.3419*****
	(0.0158)	(0.0254)	(0.0158)	(0.0251)	(0.0226)	(0.0360)	(0.0227)	(0.0361)
DivDummy	**-0.0058*****	**-0.0043***	**-0.0054*****	-0.0038	**-0.0054****	-0.0065	**-0.0055****	-0.0066
	(0.0013)	(0.0023)	(0.0013)	(0.0023)	(0.0023)	(0.0044)	(0.0023)	(0.0044)
sCFO	**0.0081*****	-0.0003	**0.0072*****	-0.0005	**0.0135****	0.0043	**0.0125****	0.0042
	(0.0031)	(0.0016)	(0.0026)	(0.0016)	(0.0062)	(0.0029)	(0.0055)	(0.0029)
SalesGrowth	**-0.0189*****	**-0.0297*****	**-0.0176*****	**-0.0287*****	**-0.0158****	**-0.0156*****	**-0.0143****	**-0.0148*****
	(0.0045)	(0.0048)	(0.0044)	(0.0048)	(0.0062)	(0.0046)	(0.0062)	(0.0047)
RDSales	**0.0089*****	-0.0018	**0.0090*****	-0.0016	**0.0152*****	-0.0004	**0.0155*****	-0.0003
	(0.0020)	(0.0015)	(0.0020)	(0.0015)	(0.0032)	(0.0024)	(0.0032)	(0.0024)
CashFlowAt	**0.1415*****	**0.1741*****	**0.1368*****	**0.1729*****	**0.2109*****	**0.1796*****	**0.2012*****	**0.1767*****
	(0.0166)	(0.0180)	(0.0165)	(0.0180)	(0.0240)	(0.0257)	(0.0241)	(0.0259)
DailyVola	**0.0001****	**-0.0001*****	**0.0001****	-0.0001	**0.0004*****	**0.0001*****	**0.0003*****	0.0001
	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0001)	(0.0000)	(0.0001)	(0.0001)
Lagged_DV	**0.8605*****	**0.6188*****	**0.8571*****	**0.6204*****	**0.7310*****	**0.2231*****	**0.7250*****	**0.2242*****
	(0.0079)	(0.0133)	(0.0080)	(0.0134)	(0.0135)	(0.0193)	(0.0138)	(0.0193)
Industry FE	Yes		Yes		Yes		Yes	
Firm FE		Yes		Yes		Yes		Yes
Year FE	Yes	Yes			Yes	Yes		
Year-Quarter FE			Yes	Yes			Yes	Yes
N	38,737	38,737	38,737	38,737	38,092	38,092	38,092	38,092
R^2	0.827	0.432	0.829	0.432	0.667	0.100	0.669	0.100
Adj. R^2	0.827	0.412	0.828	0.411	0.667	0.069	0.668	0.067

Notes: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$ (one-tailed for IVs, $\beta > 0$; two-tailed for controls). Significant coefficients in **bold**. Standard errors (in parentheses) clustered at firm level. Main sample (excludes financial and utility firms). Fiscal years 2002-2016. DWZ Eq.5 decomposition: Clarity = -CEO FE; UncRes = call-level residual. R^2 includes absorbed fixed effects (not within-R^2).

Table 16: H1.2 CEO 3-IV Decomp HFC: ClarityCEO + UncResCEO (DWZ Qtr-Exp, no-look-ahead) + UncPreCEO × Unrated Constraint

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	CashRatio				CashRatio_lead			
CEO Clarity (DWZ Qtr-Exp, c)	**-0.0114*****	**-0.0132*****	**-0.0104*****	**-0.0127*****	-0.0037	**-0.0186****	-0.0019	**-0.0187****
	(0.0035)	(0.0052)	(0.0034)	(0.0053)	(0.0054)	(0.0103)	(0.0054)	(0.0103)
CEO Residual Unc. (DWZ Qtr-Exp, c)	**0.0021***	**0.0024****	**0.0021***	**0.0022****	**0.0044****	**0.0036****	**0.0044****	**0.0036****
	(0.0013)	(0.0013)	(0.0013)	(0.0013)	(0.0020)	(0.0017)	(0.0020)	(0.0017)
CEO Pres Unc. (c)	0.0000	-0.0005	0.0000	-0.0004	0.0003	0.0009	0.0000	0.0009
	(0.0013)	(0.0013)	(0.0013)	(0.0013)	(0.0025)	(0.0022)	(0.0025)	(0.0022)
Unrated	0.0021	-0.0043	-0.0012	-0.0043	**0.0051****	-0.0073	-0.0019	-0.0074
	(0.0014)	(0.0043)	(0.0018)	(0.0043)	(0.0024)	(0.0066)	(0.0032)	(0.0066)
UncRes (Qtr-Exp) × Unrated	0.0019	0.0013	0.0019	0.0013	**0.0050***	0.0023	**0.0050***	0.0022
	(0.0025)	(0.0023)	(0.0025)	(0.0023)	(0.0038)	(0.0033)	(0.0038)	(0.0033)
UncPre × Unrated	0.0014	-0.0014	0.0017	-0.0014	-0.0001	-0.0015	0.0004	-0.0014
	(0.0026)	(0.0029)	(0.0026)	(0.0029)	(0.0046)	(0.0046)	(0.0046)	(0.0046)
Leverage	**-0.0084*****	-0.0170	**-0.0094*****	-0.0160	**-0.0210***	-0.0110	**-0.0240****	-0.0107
	(0.0031)	(0.0104)	(0.0034)	(0.0104)	(0.0108)	(0.0154)	(0.0121)	(0.0154)
lnAssets	0.0003	**-0.0157*****	**-0.0016***	**-0.0157*****	**0.0016*****	**-0.0300*****	**-0.0021***	**-0.0302*****
	(0.0003)	(0.0031)	(0.0008)	(0.0031)	(0.0006)	(0.0052)	(0.0013)	(0.0053)
TobinsQ	**0.0039*****	**0.0030****	**0.0034*****	**0.0025***	**0.0049*****	-0.0011	**0.0044*****	-0.0011
	(0.0010)	(0.0015)	(0.0010)	(0.0014)	(0.0014)	(0.0019)	(0.0015)	(0.0020)
ROA	**-0.0554*****	-0.0071	**-0.0535*****	-0.0065	**-0.1336*****	**-0.0761*****	**-0.1305*****	**-0.0773*****
	(0.0139)	(0.0145)	(0.0140)	(0.0146)	(0.0211)	(0.0249)	(0.0213)	(0.0252)
Capex	**-0.1396*****	**-0.2564*****	**-0.1439*****	**-0.2536*****	**-0.2276*****	**-0.2870*****	**-0.2378*****	**-0.2865*****
	(0.0175)	(0.0276)	(0.0176)	(0.0275)	(0.0263)	(0.0384)	(0.0262)	(0.0384)
DivDummy	**-0.0055*****	-0.0041	**-0.0056*****	-0.0041	**-0.0044***	-0.0076	**-0.0046***	-0.0077
	(0.0016)	(0.0027)	(0.0016)	(0.0026)	(0.0025)	(0.0053)	(0.0026)	(0.0053)
sCFO	**0.0099*****	0.0017	**0.0094*****	0.0017	**0.0123***	0.0041	**0.0116***	0.0040
	(0.0030)	(0.0016)	(0.0028)	(0.0016)	(0.0069)	(0.0032)	(0.0063)	(0.0032)
SalesGrowth	**-0.0128****	**-0.0267*****	**-0.0128****	**-0.0265*****	-0.0081	**-0.0127****	-0.0083	**-0.0127****
	(0.0051)	(0.0056)	(0.0051)	(0.0056)	(0.0074)	(0.0063)	(0.0075)	(0.0063)
RDSales	**0.0086*****	-0.0004	**0.0086*****	-0.0003	**0.0154*****	0.0010	**0.0154*****	0.0011
	(0.0022)	(0.0017)	(0.0022)	(0.0017)	(0.0035)	(0.0022)	(0.0035)	(0.0022)
CashFlowAt	**0.1220*****	**0.1745*****	**0.1204*****	**0.1754*****	**0.2021*****	**0.1992*****	**0.1968*****	**0.1990*****
	(0.0169)	(0.0188)	(0.0169)	(0.0189)	(0.0288)	(0.0317)	(0.0288)	(0.0317)
DailyVola	0.0000	**-0.0001****	-0.0000	**-0.0001****	**0.0003*****	0.0000	**0.0002*****	0.0000
	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0001)	(0.0000)	(0.0001)	(0.0001)
Lagged_DV	**0.8737*****	**0.6448*****	**0.8713*****	**0.6451*****	**0.7471*****	**0.2124*****	**0.7408*****	**0.2122*****
	(0.0096)	(0.0166)	(0.0096)	(0.0166)	(0.0152)	(0.0230)	(0.0154)	(0.0229)
Industry FE	Yes		Yes		Yes		Yes	
Firm FE		Yes		Yes		Yes		Yes
Year FE	Yes	Yes			Yes	Yes		
Year-Quarter FE			Yes	Yes			Yes	Yes
N	21,824	21,824	21,824	21,824	21,371	21,371	21,371	21,371
R^2	0.841	0.453	0.841	0.453	0.679	0.094	0.680	0.094
Adj. R^2	0.841	0.423	0.841	0.423	0.678	0.046	0.679	0.044

Notes: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$ (one-tailed for IVs, $\beta > 0$; two-tailed for controls). Significant coefficients in **bold**. Standard errors (in parentheses) clustered at firm level. Main sample (excludes financial and utility firms). Fiscal years 2002-2016. DWZ Eq.5 decomposition under quarterly-expanding window (no-look-ahead). R^2 includes absorbed fixed effects (not within-R^2).

Table 17: H11: Political Risk and Language Uncertainty

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
	Industry FE						Firm FE					
	UncAnsMgr	UncAnsCEO	UncPreMgr	UncPreCEO	UncAnsNoCEO	UncPreNoCEO	UncAnsMgr	UncAnsCEO	UncPreMgr	UncPreCEO	UncAnsNoCEO	UncPreNoCEO
Political Risk$_t$	**0.0002*****	**0.0001*****	**0.0002*****	**0.0003*****	**0.0001*****	**0.0001*****	**0.0001*****	**0.0001*****	**0.0001*****	**0.0002*****	**0.0001*****	**0.0001*****
	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)
UncQue	**0.0431*****	**0.0452*****	**0.0240*****	**0.0204*****	**0.0504*****	**0.0270*****	**0.0346*****	**0.0380*****	**0.0091*****	**0.0081*****	**0.0410*****	**0.0130*****
	(0.0035)	(0.0043)	(0.0041)	(0.0045)	(0.0057)	(0.0077)	(0.0026)	(0.0033)	(0.0024)	(0.0029)	(0.0051)	(0.0044)
NegCall	**0.0890*****	**0.0867*****	**0.1248*****	**0.1874*****	**0.0652*****	**0.0562****	**0.0618*****	**0.0672*****	**0.1093*****	**0.1477*****	**0.0372*****	**0.0761*****
	(0.0091)	(0.0107)	(0.0124)	(0.0141)	(0.0127)	(0.0242)	(0.0059)	(0.0075)	(0.0072)	(0.0082)	(0.0099)	(0.0130)
lnAssets	**-0.0218*****	**-0.0199*****	**-0.0333*****	**-0.0296*****	**-0.0347*****	**-0.0532*****	-0.0012	-0.0001	0.0046	0.0048	-0.0037	0.0050
	(0.0026)	(0.0031)	(0.0042)	(0.0039)	(0.0033)	(0.0073)	(0.0050)	(0.0062)	(0.0075)	(0.0078)	(0.0080)	(0.0137)
TobinsQ	**-0.0106*****	**-0.0115*****	0.0002	-0.0007	**-0.0116*****	**-0.0091***	0.0022	0.0025	0.0012	0.0001	-0.0011	-0.0037
	(0.0024)	(0.0025)	(0.0034)	(0.0036)	(0.0029)	(0.0055)	(0.0019)	(0.0026)	(0.0022)	(0.0023)	(0.0027)	(0.0037)
ROA	**0.0763*****	**0.1018*****	**0.0720****	0.0477	**0.0805*****	0.0600	**0.0572*****	**0.0604*****	0.0029	0.0346	**0.0625****	-0.0537
	(0.0236)	(0.0281)	(0.0333)	(0.0318)	(0.0292)	(0.0578)	(0.0164)	(0.0207)	(0.0217)	(0.0230)	(0.0289)	(0.0379)
CashRatio	-0.0234	0.0080	**0.0923*****	-0.0354	**-0.0642****	0.0967	**0.0581*****	**0.0463***	-0.0023	0.0265	0.0485	-0.0748
	(0.0253)	(0.0300)	(0.0344)	(0.0357)	(0.0295)	(0.0613)	(0.0189)	(0.0238)	(0.0278)	(0.0287)	(0.0335)	(0.0504)
DivDummy	0.0100	0.0121	**-0.0481*****	-0.0139	0.0016	**-0.0731*****	-0.0094	-0.0043	**-0.0185****	**-0.0229****	0.0095	-0.0302
	(0.0076)	(0.0088)	(0.0117)	(0.0118)	(0.0099)	(0.0220)	(0.0073)	(0.0095)	(0.0092)	(0.0103)	(0.0104)	(0.0184)
FirmMat	0.0012	0.0006	**0.0044***	**0.0021****	0.0016	0.0051	**0.0030*****	**0.0033*****	0.0009	0.0006	-0.0020	0.0016
	(0.0013)	(0.0017)	(0.0024)	(0.0010)	(0.0021)	(0.0045)	(0.0006)	(0.0009)	(0.0012)	(0.0007)	(0.0026)	(0.0020)
EarnVol	-0.0076	-0.0107	-0.0151	-0.0072	-0.0097	-0.0267	-0.0044	-0.0022	0.0081	-0.0104	-0.0059	0.0266
	(0.0082)	(0.0114)	(0.0179)	(0.0157)	(0.0100)	(0.0278)	(0.0076)	(0.0113)	(0.0088)	(0.0132)	(0.0093)	(0.0184)
UncPreMgr	**0.1637*****						**0.1174*****					
	(0.0097)						(0.0065)					
UncPreCEO		**0.1786*****						**0.1078*****				
		(0.0089)						(0.0056)				
UncPreNoCEO					**0.0598*****						**0.0577*****	
					(0.0091)						(0.0077)	
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes						
Firm FE							Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	77,658	65,394	77,758	65,760	73,816	76,549	77,658	65,394	77,758	65,760	73,816	76,549
R^2	0.066	0.057	0.053	0.043	0.018	0.029	0.026	0.021	0.018	0.023	0.005	0.005
Adj. R^2	0.065	0.056	0.052	0.043	0.018	0.029	0.002	-0.006	-0.005	-0.004	-0.020	-0.020

Notes: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$ (one-tailed for IVs, $\beta > 0$; two-tailed for controls). Significant coefficients in **bold**. Standard errors (in parentheses) clustered at firm level. Main sample (excludes financial and utility firms). Firms with fewer than 5 calls are excluded. R^2 includes absorbed fixed effects (not within-R^2).

Table 18: H11-Lag1: Political Risk (1-Qtr Lag) and Language Uncertainty

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
	Industry FE						Firm FE					
	UncAnsMgr	UncAnsCEO	UncPreMgr	UncPreCEO	UncAnsNoCEO	UncPreNoCEO	UncAnsMgr	UncAnsCEO	UncPreMgr	UncPreCEO	UncAnsNoCEO	UncPreNoCEO
PRisk_{t-1}	**0.0001*****	**0.0001*****	**0.0001*****	**0.0002*****	**0.0001*****	**0.0001*****	**0.0000*****	**0.0000*****	**0.0001*****	**0.0001*****	**0.0001*****	**0.0000*****
	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)
UncQue	**0.0448*****	**0.0462*****	**0.0263*****	**0.0247*****	**0.0523*****	**0.0265*****	**0.0357*****	**0.0381*****	**0.0108*****	**0.0097*****	**0.0421*****	**0.0140*****
	(0.0036)	(0.0044)	(0.0042)	(0.0046)	(0.0057)	(0.0079)	(0.0026)	(0.0034)	(0.0025)	(0.0030)	(0.0051)	(0.0044)
NegCall	**0.0935*****	**0.0919*****	**0.1342*****	**0.1981*****	**0.0686*****	**0.0634*****	**0.0668*****	**0.0715*****	**0.1170*****	**0.1591*****	**0.0417*****	**0.0834*****
	(0.0092)	(0.0109)	(0.0126)	(0.0144)	(0.0128)	(0.0244)	(0.0060)	(0.0077)	(0.0073)	(0.0084)	(0.0101)	(0.0131)
lnAssets	**-0.0217*****	**-0.0191*****	**-0.0335*****	**-0.0298*****	**-0.0352*****	**-0.0527*****	-0.0014	0.0009	0.0045	0.0054	-0.0064	0.0051
	(0.0026)	(0.0031)	(0.0043)	(0.0040)	(0.0033)	(0.0074)	(0.0052)	(0.0065)	(0.0077)	(0.0080)	(0.0081)	(0.0141)
TobinsQ	**-0.0108*****	**-0.0119*****	0.0005	-0.0003	**-0.0115*****	-0.0085	0.0020	0.0021	0.0014	0.0001	-0.0008	-0.0038
	(0.0024)	(0.0025)	(0.0034)	(0.0036)	(0.0030)	(0.0055)	(0.0020)	(0.0027)	(0.0023)	(0.0023)	(0.0028)	(0.0037)
ROA	**0.0885*****	**0.1151*****	**0.0718****	0.0480	**0.0854*****	0.0442	**0.0610*****	**0.0633*****	0.0029	**0.0455***	**0.0614****	-0.0592
	(0.0249)	(0.0295)	(0.0335)	(0.0331)	(0.0303)	(0.0589)	(0.0175)	(0.0219)	(0.0229)	(0.0246)	(0.0294)	(0.0395)
CashRatio	-0.0217	0.0126	**0.0918*****	-0.0347	**-0.0647****	0.0958	**0.0598*****	**0.0534****	-0.0016	0.0271	0.0334	-0.0647
	(0.0257)	(0.0305)	(0.0352)	(0.0365)	(0.0301)	(0.0620)	(0.0192)	(0.0243)	(0.0284)	(0.0290)	(0.0333)	(0.0509)
DivDummy	0.0102	0.0117	**-0.0501*****	-0.0145	0.0018	**-0.0725*****	-0.0097	-0.0061	**-0.0183****	**-0.0242****	0.0113	-0.0288
	(0.0077)	(0.0088)	(0.0119)	(0.0120)	(0.0100)	(0.0220)	(0.0074)	(0.0097)	(0.0093)	(0.0107)	(0.0107)	(0.0183)
FirmMat	0.0003	-0.0004	**0.0059*****	0.0021	0.0014	**0.0090****	**0.0028***	**0.0039****	0.0017	-0.0007	-0.0017	0.0047
	(0.0017)	(0.0022)	(0.0023)	(0.0014)	(0.0021)	(0.0038)	(0.0015)	(0.0019)	(0.0024)	(0.0015)	(0.0027)	(0.0035)
EarnVol	-0.0065	-0.0074	-0.0174	-0.0070	-0.0077	-0.0283	-0.0039	0.0035	0.0082	-0.0131	-0.0047	0.0341
	(0.0106)	(0.0125)	(0.0201)	(0.0158)	(0.0137)	(0.0325)	(0.0082)	(0.0115)	(0.0107)	(0.0146)	(0.0118)	(0.0230)
UncPreMgr	**0.1693*****						**0.1219*****					
	(0.0099)						(0.0066)					
UncPreCEO		**0.1821*****						**0.1122*****				
		(0.0090)						(0.0058)				
UncPreNoCEO					**0.0614*****						**0.0583*****	
					(0.0092)						(0.0078)	
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes						
Firm FE							Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	74,918	63,049	75,014	63,399	71,264	73,875	74,918	63,049	75,014	63,399	71,264	73,875
R^2	0.063	0.055	0.049	0.036	0.018	0.028	0.022	0.018	0.014	0.016	0.005	0.004
Adj. R^2	0.063	0.054	0.049	0.035	0.018	0.027	-0.002	-0.009	-0.011	-0.012	-0.021	-0.021

Notes: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$ (one-tailed for IVs, $\beta > 0$; two-tailed for controls). Significant coefficients in **bold**. Standard errors (in parentheses) clustered at firm level. Main sample (excludes financial and utility firms). Firms with fewer than 5 calls are excluded. R^2 includes absorbed fixed effects (not within-R^2).

Table 19: H11-Lag2: Political Risk (2-Qtr Lag) and Language Uncertainty

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
	Industry FE						Firm FE					
	UncAnsMgr	UncAnsCEO	UncPreMgr	UncPreCEO	UncAnsNoCEO	UncPreNoCEO	UncAnsMgr	UncAnsCEO	UncPreMgr	UncPreCEO	UncAnsNoCEO	UncPreNoCEO
$PRisk_{t-2}$	**0.0001*****	**0.0001*****	**0.0001*****	**0.0001*****	**0.0001*****	**0.0001****	**0.0000*****	**0.0000****	**0.0000*****	**0.0000*****	**0.0000****	0.0000
	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)
UncQue	**0.0447*****	**0.0470*****	**0.0265*****	**0.0242*****	**0.0519*****	**0.0287*****	**0.0358*****	**0.0395*****	**0.0108*****	**0.0099*****	**0.0419*****	**0.0146*****
	(0.0036)	(0.0044)	(0.0043)	(0.0047)	(0.0058)	(0.0080)	(0.0027)	(0.0034)	(0.0025)	(0.0031)	(0.0052)	(0.0045)
NegCall	**0.0937*****	**0.0923*****	**0.1365*****	**0.2019*****	**0.0688*****	**0.0654*****	**0.0657*****	**0.0719*****	**0.1180*****	**0.1609*****	**0.0400*****	**0.0826*****
	(0.0092)	(0.0109)	(0.0127)	(0.0145)	(0.0129)	(0.0244)	(0.0060)	(0.0077)	(0.0073)	(0.0084)	(0.0101)	(0.0131)
lnAssets	**-0.0215*****	**-0.0194*****	**-0.0332*****	**-0.0298*****	**-0.0346*****	**-0.0529*****	-0.0010	0.0016	0.0045	0.0047	-0.0050	0.0032
	(0.0026)	(0.0031)	(0.0043)	(0.0040)	(0.0034)	(0.0075)	(0.0052)	(0.0066)	(0.0076)	(0.0078)	(0.0082)	(0.0143)
TobinsQ	**-0.0105*****	**-0.0115*****	0.0004	-0.0002	**-0.0112*****	-0.0091	0.0021	0.0025	0.0014	-0.0002	-0.0008	-0.0035
	(0.0024)	(0.0026)	(0.0035)	(0.0037)	(0.0030)	(0.0055)	(0.0020)	(0.0028)	(0.0023)	(0.0023)	(0.0028)	(0.0038)
ROA	**0.0838*****	**0.1100*****	**0.0735****	0.0518	**0.0810*****	0.0477	**0.0575*****	**0.0617*****	0.0055	**0.0464***	**0.0599****	-0.0559
	(0.0247)	(0.0293)	(0.0335)	(0.0331)	(0.0308)	(0.0588)	(0.0176)	(0.0219)	(0.0221)	(0.0247)	(0.0304)	(0.0383)
CashRatio	-0.0210	0.0111	**0.0928*****	-0.0400	**-0.0648****	**0.1037***	**0.0627*****	**0.0498****	-0.0033	0.0233	0.0498	-0.0683
	(0.0259)	(0.0308)	(0.0353)	(0.0368)	(0.0304)	(0.0624)	(0.0193)	(0.0244)	(0.0281)	(0.0292)	(0.0342)	(0.0515)
DivDummy	0.0091	0.0106	**-0.0497*****	-0.0142	0.0013	**-0.0715*****	-0.0119	-0.0079	**-0.0169***	**-0.0207***	0.0092	-0.0276
	(0.0077)	(0.0089)	(0.0120)	(0.0121)	(0.0101)	(0.0221)	(0.0074)	(0.0098)	(0.0094)	(0.0106)	(0.0106)	(0.0187)
FirmMat	0.0002	-0.0005	**0.0059****	0.0021	0.0015	**0.0088****	**0.0029***	**0.0037****	0.0017	-0.0007	-0.0014	0.0050
	(0.0017)	(0.0021)	(0.0023)	(0.0015)	(0.0021)	(0.0039)	(0.0015)	(0.0018)	(0.0023)	(0.0016)	(0.0028)	(0.0033)
EarnVol	-0.0079	-0.0086	-0.0152	-0.0078	-0.0066	-0.0235	-0.0069	0.0005	0.0092	-0.0154	-0.0052	0.0376
	(0.0114)	(0.0134)	(0.0208)	(0.0159)	(0.0136)	(0.0344)	(0.0098)	(0.0129)	(0.0109)	(0.0158)	(0.0129)	(0.0234)
UncPreMgr	**0.1707*****						**0.1247*****					
	(0.0100)						(0.0067)					
UncPreCEO		**0.1827*****						**0.1138*****				
		(0.0091)						(0.0058)				
UncPreNoCEO					**0.0608*****						**0.0567*****	
					(0.0093)						(0.0078)	
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes						
Firm FE							Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	74,467	62,674	74,561	63,022	70,842	73,432	74,467	62,674	74,561	63,022	70,842	73,432
R^2	0.063	0.055	0.048	0.035	0.017	0.028	0.023	0.019	0.013	0.015	0.004	0.004
Adj. R^2	0.063	0.054	0.047	0.034	0.017	0.027	-0.002	-0.009	-0.012	-0.013	-0.022	-0.022

Notes: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$ (one-tailed for IVs, $\beta > 0$; two-tailed for controls). Significant coefficients in **bold**. Standard errors (in parentheses) clustered at firm level. Main sample (excludes financial and utility firms). Firms with fewer than 5 calls are excluded. R^2 includes absorbed fixed effects (not within-R^2).

Table 20: H23: Product-Market Competition and Uncertainty Language

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
	Industry FE						Firm FE					
	UncAnsMgr	UncAnsCEO	UncPreMgr	UncPreCEO	UncAnsNoCEO	UncPreNoCEO	UncAnsMgr	UncAnsCEO	UncPreMgr	UncPreCEO	UncAnsNoCEO	UncPreNoCEO
$z(\log(\text{TSIMM}))$	**0.0090****	0.0054	**0.0297*****	**0.0304*****	**0.0132****	0.0089	-0.0012	-0.0061	0.0058	**0.0302*****	0.0018	-0.0239
	(0.0048)	(0.0055)	(0.0075)	(0.0074)	(0.0062)	(0.0124)	(0.0059)	(0.0090)	(0.0091)	(0.0093)	(0.0095)	(0.0171)
UncQue	**0.0616*****	**0.0625*****	**0.0483*****	**0.0398*****	**0.0696*****	**0.0548*****	**0.0420*****	**0.0459*****	**0.0149****	0.0084	**0.0453*****	**0.0250***
	(0.0078)	(0.0093)	(0.0094)	(0.0100)	(0.0115)	(0.0192)	(0.0061)	(0.0078)	(0.0062)	(0.0073)	(0.0113)	(0.0128)
NegCall	**0.1109*****	**0.0900*****	**0.1553*****	**0.2340*****	**0.0928*****	**0.0669***	**0.0690*****	**0.0648*****	**0.1370*****	**0.1698*****	**0.0508*****	**0.1069*****
	(0.0133)	(0.0153)	(0.0179)	(0.0201)	(0.0187)	(0.0361)	(0.0097)	(0.0123)	(0.0120)	(0.0140)	(0.0166)	(0.0232)
lnAssets	**-0.0207*****	**-0.0177*****	**-0.0374*****	**-0.0335*****	**-0.0349*****	**-0.0549*****	0.0002	0.0007	0.0060	0.0013	0.0011	0.0138
	(0.0027)	(0.0031)	(0.0043)	(0.0038)	(0.0034)	(0.0078)	(0.0054)	(0.0067)	(0.0079)	(0.0081)	(0.0086)	(0.0140)
TobinsQ	**-0.0080*****	**-0.0089*****	-0.0014	-0.0007	**-0.0102*****	**-0.0110***	0.0013	0.0032	0.0011	-0.0017	-0.0037	-0.0016
	(0.0024)	(0.0026)	(0.0034)	(0.0037)	(0.0029)	(0.0056)	(0.0020)	(0.0028)	(0.0024)	(0.0025)	(0.0030)	(0.0040)
ROA	**0.0808*****	**0.0778*****	**0.1163*****	**0.0681****	**0.1069*****	0.0952	**0.0601*****	**0.0405***	0.0064	0.0241	**0.0797****	-0.0610
	(0.0230)	(0.0278)	(0.0322)	(0.0314)	(0.0313)	(0.0595)	(0.0184)	(0.0234)	(0.0240)	(0.0254)	(0.0317)	(0.0420)
CashRatio	-0.0339	0.0120	0.0503	**-0.0803****	**-0.0785****	0.0881	**0.0589*****	**0.0524***	-0.0130	0.0022	0.0637	-0.0776
	(0.0262)	(0.0296)	(0.0362)	(0.0360)	(0.0321)	(0.0633)	(0.0222)	(0.0268)	(0.0292)	(0.0306)	(0.0398)	(0.0533)
DivDummy	0.0110	0.0107	**-0.0425*****	-0.0060	0.0000	**-0.0772*****	-0.0068	0.0020	**-0.0163***	-0.0166	0.0135	**-0.0355***
	(0.0078)	(0.0088)	(0.0119)	(0.0117)	(0.0105)	(0.0221)	(0.0073)	(0.0101)	(0.0095)	(0.0106)	(0.0111)	(0.0190)
FirmMat	**0.0019*****	**0.0014***	0.0021	**0.0014****	-0.0001	0.0007	**0.0027*****	**0.0028*****	0.0002	0.0003	-0.0030	0.0003
	(0.0006)	(0.0008)	(0.0018)	(0.0007)	(0.0030)	(0.0031)	(0.0003)	(0.0006)	(0.0006)	(0.0004)	(0.0020)	(0.0011)
EarnVol	-0.0105	-0.0181	-0.0106	-0.0210	-0.0107	-0.0156	-0.0020	-0.0079	0.0088	-0.0114	-0.0031	0.0222
	(0.0091)	(0.0126)	(0.0145)	(0.0152)	(0.0106)	(0.0209)	(0.0072)	(0.0131)	(0.0073)	(0.0146)	(0.0086)	(0.0175)
UncPreMgr	**0.1943*****						**0.1418*****					
	(0.0128)						(0.0104)					
UncPreCEO		**0.2491*****						**0.1687*****				
		(0.0138)						(0.0108)				
UncPreNoCEO					**0.0489*****						**0.0478*****	
					(0.0114)						(0.0115)	
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes						
Firm FE							Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	20,768	18,447	20,774	18,492	20,236	20,513	20,768	18,447	20,774	18,492	20,236	20,513
R^2	0.104	0.103	0.064	0.050	0.029	0.035	0.038	0.038	0.017	0.021	0.006	0.006
Adj. R^2	0.102	0.102	0.062	0.048	0.028	0.033	-0.045	-0.050	-0.067	-0.068	-0.081	-0.080

Notes: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$ (one-tailed for IVs, $\beta > 0$; two-tailed for controls). Significant coefficients in **bold**. Standard errors (in parentheses) clustered at firm level. Main sample (excludes financial and utility firms). Unit of observation: firm-fiscal-year. R^2 includes absorbed fixed effects (not within-R^2).

Table 21: H14c CEO 3-IV Decomp: ClarityCEO + UncResCEO (DWZ Eq.5 Full) + UncPreCEO (raw) and 25-Day Post-Call Bid-Ask Spread Level ($[0, +25]$, the call day plus 25 trading days)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
	$\text{Spread}_{25D,t}$						$\text{Spread}_{25D,t+1}$					
CEO Clarity (DWZ)	-0.0477	1.0617	0.2122	1.1056	0.1644	1.0547	-0.1886	0.5998	-0.0922	0.6828	-0.0431	0.5722
	(0.2145)	(0.4277)	(0.2227)	(0.4260)	(0.2103)	(0.4174)	(0.2370)	(0.4004)	(0.2399)	(0.4055)	(0.2206)	(0.3729)
CEO Residual Uncertainty (DWZ)	-0.0594	-0.0687	-0.0882	-0.1021	-0.0757	-0.0893	0.0825	0.0696	0.0742	0.0545	0.0984	0.0751
	(0.1068)	(0.1057)	(0.1023)	(0.1007)	(0.0994)	(0.0981)	(0.1416)	(0.1437)	(0.1407)	(0.1426)	(0.1339)	(0.1357)
CEO Pres Uncertainty	**0.1664****	**0.3496*****	0.0814	**0.2945*****	0.0108	**0.1644***	-0.0817	-0.0288	-0.1072	-0.0360	0.0170	0.1124
	(0.0946)	(0.1253)	(0.0956)	(0.1229)	(0.0919)	(0.1184)	(0.0926)	(0.1229)	(0.0926)	(0.1233)	(0.0872)	(0.1186)
lnAssets	**-0.8039*****	**-1.9098*****	**-0.6339*****	**-1.6825*****	**-0.6352*****	**-1.8277*****	**-0.8218*****	**-1.5191*****	**-0.7857*****	**-1.7292*****	**-0.6568*****	**-1.4181*****
	(0.0490)	(0.2000)	(0.0472)	(0.2156)	(0.0473)	(0.2158)	(0.0513)	(0.1951)	(0.0505)	(0.2130)	(0.0474)	(0.2017)
TobinsQ	**-0.2861*****	**-0.4785*****	**-0.2633*****	**-0.5518*****	**-0.2654*****	**-0.5883*****	**-0.2305*****	**-0.3215*****	**-0.2285*****	**-0.4803*****	**-0.2035*****	**-0.4289*****
	(0.0457)	(0.0900)	(0.0470)	(0.0963)	(0.0451)	(0.0956)	(0.0467)	(0.0850)	(0.0478)	(0.1002)	(0.0433)	(0.0935)
ROA	**-9.4469*****	**-11.6430*****	**-5.7059*****	**-7.6844*****	**-5.7511*****	**-7.8517*****	**-9.9256*****	**-12.8443*****	**-8.8744*****	**-11.9197*****	**-7.6375*****	**-10.5867*****
	(1.0201)	(1.3867)	(0.9822)	(1.2654)	(0.9633)	(1.2490)	(1.0798)	(1.2992)	(1.0722)	(1.2732)	(0.9869)	(1.1921)
Leverage	**0.3875***	**1.8330*****	-0.1126	0.6850	-0.0542	0.8026	0.2924	0.5282	0.1338	0.2221	0.0592	0.1928
	(0.2088)	(0.5143)	(0.2434)	(0.5013)	(0.2264)	(0.4954)	(0.2224)	(0.5516)	(0.2248)	(0.5596)	(0.2111)	(0.5258)
Capex	-0.8001	0.9540	**-2.3622****	-1.5271	**-2.0790***	-1.2897	**-1.6463***	-3.2121	**-2.1561****	**-4.3379***	-1.4770	-2.4356
	(1.1109)	(2.9126)	(1.1044)	(2.8061)	(1.0845)	(2.7998)	(0.9620)	(2.2020)	(0.9823)	(2.2446)	(0.9011)	(2.1565)
DivDummy	0.1043	0.1429	**0.3329*****	0.2549	**0.3156*****	0.2723	0.0240	0.1007	0.1012	0.1454	0.1145	0.1739
	(0.0893)	(0.2020)	(0.0896)	(0.1888)	(0.0855)	(0.1867)	(0.0887)	(0.2095)	(0.0889)	(0.2074)	(0.0825)	(0.1975)
sCFO	**0.6700****	**0.8405****	0.4045	**0.6825***	0.4084	**0.6882****	0.0871	**0.3644***	0.0040	0.3165	-0.0840	0.2479
	(0.3405)	(0.3472)	(0.4604)	(0.3789)	(0.4126)	(0.3375)	(0.1759)	(0.1877)	(0.2174)	(0.2055)	(0.1937)	(0.1738)
Lagged_DV	**0.7547*****	**0.6569*****	**0.7140*****	**0.6180*****	**0.7310*****	**0.6318*****	**0.7540*****	**0.6361*****	**0.7360*****	**0.6135*****	**0.7676*****	**0.6518*****
	(0.0150)	(0.0186)	(0.0167)	(0.0199)	(0.0168)	(0.0208)	(0.0156)	(0.0185)	(0.0173)	(0.0205)	(0.0162)	(0.0197)
StockPrice			-0.0011	-0.0033	0.0002	0.0019			-0.0001	**0.0101*****	-0.0016	**0.0047****
			(0.0014)	(0.0024)	(0.0013)	(0.0024)			(0.0013)	(0.0024)	(0.0012)	(0.0022)
Turnover			**-24.1012*****	**-23.8647*****	**-20.7840*****	**-20.3411*****			**-7.9396*****	**-5.5050*****	**-7.1303*****	**-4.7073****
			(1.7508)	(2.2119)	(1.7290)	(2.1791)			(1.5411)	(2.0904)	(1.4699)	(1.9633)
DailyVola			**0.1229*****	**0.1393*****	**0.1075*****	**0.1257*****			**0.0404*****	**0.0520*****	**0.0396*****	**0.0441*****
			(0.0072)	(0.0081)	(0.0082)	(0.0099)			(0.0062)	(0.0068)	(0.0069)	(0.0081)
AbsSurpDec			**-0.0919*****	-0.0241	**-0.0756*****	-0.0093			-0.0050	0.0119	-0.0174	-0.0012
			(0.0265)	(0.0300)	(0.0253)	(0.0286)			(0.0278)	(0.0304)	(0.0261)	(0.0285)
Extended Controls			Yes	Yes	Yes	Yes			Yes	Yes	Yes	Yes
Industry FE	Yes		Yes		Yes		Yes		Yes		Yes	
Firm FE		Yes		Yes		Yes		Yes		Yes		Yes
Year FE	Yes	Yes	Yes	Yes			Yes	Yes	Yes	Yes		
Year-Quarter FE					Yes	Yes					Yes	Yes
N	42,625	42,625	42,625	42,625	42,625	42,625	43,049	43,049	43,049	43,049	43,049	43,049
R^2	0.703	0.517	0.721	0.551	0.730	0.548	0.700	0.499	0.702	0.504	0.723	0.523
Adj. R^2	0.703	0.501	0.721	0.535	0.730	0.532	0.700	0.482	0.702	0.487	0.723	0.506

Notes: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$ (one-tailed for IVs, $\beta > 0$; two-tailed for controls). Significant coefficients in **bold**. Standard errors (in parentheses) clustered at firm level. Main sample (excludes financial and utility firms). DWZ Full decomposition: Clarity = -CEO FE; UncRes = call-level residual. R^2 includes absorbed fixed effects (not within-R^2). Spread_{25D} values rescaled by 10^4 (basis points) for readability; multiply coefficients by 10^{-4} to recover raw fraction units.

Table 22: H5 CEO 3-IV Decomp: ClarityCEO + UncResCEO (DWZ Eq.5 Full) + UncPreCEO (raw) and Analyst Forecast Dispersion (Wang 2020)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
	$DISP_t$						$DISP_{t+1}$					
CEO Clarity (DWZ)	**-0.0002***	0.0005	**-0.0001***	0.0005	**-0.0001***	0.0005	-0.0001	0.0007	-0.0001	0.0007	-0.0001	0.0007
	(0.0001)	(0.0003)	(0.0001)	(0.0002)	(0.0001)	(0.0002)	(0.0001)	(0.0002)	(0.0001)	(0.0002)	(0.0001)	(0.0002)
CEO Residual Uncertainty (DWZ)	-0.0001	-0.0001	-0.0001	-0.0001	-0.0001	-0.0001	0.0000	-0.0000	0.0000	-0.0000	0.0000	-0.0000
	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0001)	(0.0001)	(0.0000)	(0.0001)	(0.0000)	(0.0001)
CEO Pres Uncertainty	**0.0001****	**0.0001***	0.0000	0.0000	0.0000	-0.0000	**0.0001****	**0.0001***	0.0000	0.0000	0.0001	0.0000
	(0.0000)	(0.0001)	(0.0000)	(0.0001)	(0.0000)	(0.0001)	(0.0000)	(0.0001)	(0.0000)	(0.0001)	(0.0000)	(0.0001)
lnAssets	-0.0000	-0.0001	-0.0000	-0.0001	-0.0000	-0.0001	**-0.0000***	-0.0001	**-0.0000***	-0.0001	-0.0000	**-0.0001***
	(0.0000)	(0.0001)	(0.0000)	(0.0001)	(0.0000)	(0.0001)	(0.0000)	(0.0001)	(0.0000)	(0.0001)	(0.0000)	(0.0001)
TobinsQ	**-0.0001*****	**-0.0001****	**-0.0001*****	**-0.0001****	**-0.0001*****	**-0.0001****	**-0.0001*****	**-0.0001*****	**-0.0001*****	**-0.0001*****	**-0.0001*****	**-0.0001*****
	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)
ROA	**-0.0027*****	**-0.0039*****	**-0.0010****	**-0.0024*****	**-0.0011*****	**-0.0025*****	**-0.0023*****	**-0.0037*****	**-0.0011****	**-0.0026*****	**-0.0010****	**-0.0026*****
	(0.0004)	(0.0006)	(0.0004)	(0.0006)	(0.0004)	(0.0006)	(0.0004)	(0.0006)	(0.0004)	(0.0006)	(0.0004)	(0.0006)
Leverage	**0.0004*****	**0.0014*****	**0.0004*****	**0.0013*****	**0.0004*****	**0.0013*****	**0.0004*****	**0.0011*****	**0.0004*****	**0.0011*****	**0.0004*****	**0.0010*****
	(0.0001)	(0.0003)	(0.0001)	(0.0003)	(0.0001)	(0.0003)	(0.0001)	(0.0003)	(0.0001)	(0.0003)	(0.0001)	(0.0003)
Capex	-0.0006	**-0.0017****	-0.0004	**-0.0017****	-0.0004	**-0.0018****	-0.0004	-0.0010	-0.0002	-0.0009	-0.0003	-0.0012
	(0.0006)	(0.0008)	(0.0006)	(0.0008)	(0.0006)	(0.0008)	(0.0006)	(0.0008)	(0.0006)	(0.0008)	(0.0006)	(0.0008)
DivDummy	**-0.0002*****	-0.0001	**-0.0002*****	-0.0001	**-0.0002*****	-0.0001	**-0.0002*****	-0.0001	**-0.0002*****	-0.0001	**-0.0002*****	-0.0001
	(0.0000)	(0.0001)	(0.0000)	(0.0001)	(0.0000)	(0.0001)	(0.0000)	(0.0001)	(0.0000)	(0.0001)	(0.0000)	(0.0001)
sCFO	**0.0005***	**0.0002****	**0.0005****	**0.0002****	**0.0005****	**0.0002****	**0.0005***	**0.0001***	**0.0005***	**0.0001***	**0.0005***	**0.0001***
	(0.0003)	(0.0001)	(0.0003)	(0.0001)	(0.0003)	(0.0001)	(0.0003)	(0.0001)	(0.0003)	(0.0001)	(0.0003)	(0.0001)
Lagged_DV	**0.6498*****	**0.3798*****	**0.6215*****	**0.3598*****	**0.6257*****	**0.3635*****	**0.6429*****	**0.3669*****	**0.6171*****	**0.3459*****	**0.6199*****	**0.3489*****
	(0.0186)	(0.0280)	(0.0185)	(0.0264)	(0.0184)	(0.0264)	(0.0188)	(0.0284)	(0.0190)	(0.0270)	(0.0189)	(0.0270)
SurpDec			**-0.0000*****	-0.0000	**-0.0000*****	-0.0000			-0.0000	-0.0000	-0.0000	-0.0000
			(0.0000)	(0.0000)	(0.0000)	(0.0000)			(0.0000)	(0.0000)	(0.0000)	(0.0000)
Loss			**0.0008*****	**0.0006*****	**0.0008*****	**0.0006*****			**0.0007*****	**0.0005*****	**0.0007*****	**0.0005*****
			(0.0001)	(0.0001)	(0.0001)	(0.0001)			(0.0001)	(0.0001)	(0.0001)	(0.0001)
UncQue			0.0000	0.0000	0.0000	0.0000			0.0000	0.0000	0.0000	0.0000
			(0.0000)	(0.0000)	(0.0000)	(0.0000)			(0.0000)	(0.0000)	(0.0000)	(0.0000)
NegCall			**0.0006*****	**0.0008*****	**0.0005*****	**0.0008*****			**0.0004*****	**0.0007*****	**0.0003*****	**0.0006*****
			(0.0001)	(0.0001)	(0.0001)	(0.0001)			(0.0001)	(0.0001)	(0.0001)	(0.0001)
Extended Controls			Yes	Yes	Yes	Yes			Yes	Yes	Yes	Yes
Industry FE	Yes		Yes		Yes		Yes		Yes		Yes	
Firm FE		Yes		Yes		Yes		Yes		Yes		Yes
Year FE	Yes	Yes	Yes	Yes			Yes	Yes	Yes	Yes		
Year-Quarter FE					Yes	Yes					Yes	Yes
N	17,908	17,908	17,908	17,908	17,908	17,908	18,287	18,287	18,287	18,287	18,287	18,287
R^2	0.485	0.198	0.497	0.216	0.499	0.216	0.481	0.192	0.488	0.203	0.492	0.203
Adj. R^2	0.484	0.155	0.496	0.174	0.497	0.171	0.480	0.149	0.487	0.161	0.489	0.158

Notes: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$ (one-tailed for IVs, $\beta > 0$; two-tailed for controls). Significant coefficients in **bold**. Standard errors (in parentheses) clustered at firm level. Main sample (excludes financial and utility firms). DWZ Full decomposition: Clarity = -CEO FE; UncRes = call-level residual. R^2 includes absorbed fixed effects (not within-R^2).

Table 23: Consolidated Driver Matrix: Exogenous Uncertainty Drivers of CEO Speech Uncertainty

	UncAnsCEO		UncPreCEO	
	Industry FE	Firm FE	Industry FE	Firm FE
PRisk (firm-call, Hassan et al. (2020))	**0.00014*****	**0.00013*****	**0.00030*****	**0.00022*****
	(0.00002)	(0.00001)	(0.00002)	(0.00001)
US EPU (log, Baker et al. (2016))	**0.0117***	**0.0138****	**0.0211*****	**0.0239*****
	(0.0072)	(0.0077)	(0.0074)	(0.0076)
Global EPU (log, Davis (2016))	**0.0178***	**0.0212****	**0.0271*****	**0.0309*****
	(0.0109)	(0.0110)	(0.0107)	(0.0108)
N range	59,676–65,394	59,676–65,394	60,503–65,760	60,503–65,760

Notes: Each cell reports the coefficient and cluster-robust standard error (in parentheses) on the respective driver variable from a PanelOLS regression of the CEO speech-uncertainty regressand (UncAnsCEO or UncPreCEO) on the driver plus the standard control set. PRisk is the firm-call-level political-risk count of Hassan et al. (2020); US EPU is the news-based US Economic Policy Uncertainty index of Baker et al. (2016) (natural log, calendar-monthly match); Global EPU is the GDP-weighted global extension of Davis (2016) (natural log, calendar-monthly match). Industry FE uses Fama-French 12 dummies; firm FE uses `gvkey` entity effects; calendar-year FE absorbed in all specifications. Standard errors clustered at firm level (PRisk, suite H11); two-way firm × calendar-year-quarter (US EPU and Global EPU, suites H24 and H24b). $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$ (one-tailed positive on all drivers). Significant coefficients in **bold**.

Table 24: H24b: Global Economic Policy Uncertainty and Call Language Uncertainty

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
	Industry + Cal. Year FE						Firm + Cal. Year FE					
	UncAnsMgr	UncPreMgr	UncAnsCEO	UncPreCEO	UncAnsNoCEO	UncPreNoCEO	UncAnsMgr	UncPreMgr	UncAnsCEO	UncPreCEO	UncAnsNoCEO	UncPreNoCEO
$\log(\text{GEPU})_t$	**0.0234*****	**0.0131***	**0.0178***	**0.0271*****	**0.0178***	0.0078	**0.0242****	**0.0124***	**0.0212****	**0.0309*****	0.0166	0.0053
	(0.0099)	(0.0093)	(0.0109)	(0.0107)	(0.0128)	(0.0123)	(0.0104)	(0.0086)	(0.0110)	(0.0108)	(0.0132)	(0.0108)
UncQue	**0.0388*****	**0.0115*****	**0.0413*****	**0.0164*****	**0.0483*****	**0.0114*****	**0.0348*****	**0.0084*****	**0.0380*****	**0.0095*****	**0.0409*****	**0.0103*****
	(0.0032)	(0.0023)	(0.0035)	(0.0032)	(0.0062)	(0.0037)	(0.0029)	(0.0023)	(0.0031)	(0.0028)	(0.0059)	(0.0039)
NegCall	**0.0704*****	**0.0724*****	**0.0669*****	**0.1247*****	**0.0617*****	**0.0352*****	**0.0597*****	**0.0926*****	**0.0619*****	**0.1385*****	**0.0374*****	**0.0608*****
	(0.0065)	(0.0058)	(0.0083)	(0.0089)	(0.0124)	(0.0077)	(0.0054)	(0.0058)	(0.0067)	(0.0079)	(0.0115)	(0.0085)
lnAssets	**-0.0153*****	**-0.0127*****	**-0.0136*****	**-0.0143*****	**-0.0297*****	**-0.0143*****	-0.0005	0.0027	0.0011	0.0037	-0.0042	0.0037
	(0.0018)	(0.0016)	(0.0022)	(0.0020)	(0.0027)	(0.0023)	(0.0045)	(0.0046)	(0.0063)	(0.0063)	(0.0083)	(0.0066)
TobinsQ	**-0.0071*****	0.0011	**-0.0081*****	0.0008	**-0.0097*****	-0.0017	0.0023	0.0015	0.0025	0.0007	0.0006	-0.0017
	(0.0016)	(0.0012)	(0.0018)	(0.0019)	(0.0025)	(0.0015)	(0.0017)	(0.0014)	(0.0024)	(0.0018)	(0.0027)	(0.0021)
ROA	**0.0606*****	**0.0342*****	**0.0836*****	**0.0367****	**0.0632*****	0.0228	**0.0504*****	0.0122	**0.0547*****	**0.0462****	0.0311	-0.0079
	(0.0170)	(0.0122)	(0.0223)	(0.0168)	(0.0238)	(0.0169)	(0.0160)	(0.0145)	(0.0193)	(0.0191)	(0.0252)	(0.0203)
CashRatio	-0.0120	**0.0325*****	0.0184	-0.0080	**-0.0519****	0.0209	**0.0536*****	0.0062	**0.0605*****	0.0304	0.0429	-0.0195
	(0.0174)	(0.0119)	(0.0220)	(0.0188)	(0.0251)	(0.0160)	(0.0149)	(0.0161)	(0.0226)	(0.0231)	(0.0293)	(0.0247)
DivDummy	0.0074	**-0.0167*****	0.0104	-0.0074	0.0047	**-0.0165*****	-0.0079	**-0.0101***	-0.0034	**-0.0156***	0.0109	-0.0130
	(0.0053)	(0.0042)	(0.0064)	(0.0061)	(0.0078)	(0.0058)	(0.0066)	(0.0059)	(0.0089)	(0.0083)	(0.0095)	(0.0092)
FirmMat	-0.0002	**0.0021*****	-0.0005	0.0009	0.0006	**0.0021****	**0.0027***	0.0008	**0.0039****	-0.0011	-0.0020	0.0013
	(0.0011)	(0.0007)	(0.0015)	(0.0008)	(0.0019)	(0.0009)	(0.0015)	(0.0013)	(0.0019)	(0.0012)	(0.0028)	(0.0011)
EarnVol	-0.0060	-0.0061	-0.0002	-0.0074	-0.0168	-0.0033	-0.0029	0.0021	0.0150	-0.0082	-0.0160	0.0134
	(0.0075)	(0.0069)	(0.0106)	(0.0086)	(0.0112)	(0.0092)	(0.0076)	(0.0067)	(0.0111)	(0.0119)	(0.0122)	(0.0113)
UncPreMgr	**0.1223*****						**0.1126*****					
	(0.0071)						(0.0059)					
UncPreCEO			**0.1384*****						**0.1043*****			
			(0.0070)						(0.0049)			
UncPreNoCEO					**0.0563*****						**0.0614*****	
					(0.0077)						(0.0071)	
Lagged_DV	**0.3334*****	**0.6755*****	**0.2986*****	**0.5176*****	**0.1937*****	**0.7610*****	**0.1368*****	**0.4185*****	**0.1103*****	**0.2626*****	**0.0762*****	**0.5359*****
	(0.0084)	(0.0101)	(0.0091)	(0.0131)	(0.0068)	(0.0107)	(0.0069)	(0.0099)	(0.0071)	(0.0089)	(0.0064)	(0.0133)
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes						
Firm FE							Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	74,013	75,142	59,676	60,503	68,858	73,835	74,013	75,142	59,676	60,503	68,858	73,835
R^2	0.171	0.484	0.141	0.292	0.056	0.592	0.041	0.186	0.031	0.084	0.010	0.291
Adj. R^2	0.171	0.484	0.141	0.291	0.055	0.592	0.017	0.166	0.002	0.057	-0.016	0.273

Notes: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$ (one-tailed for IVs, $\beta > 0$; two-tailed for controls). Significant coefficients in **bold**. Standard errors (in parentheses) two-way clustered (firm, calendar quarter). Main sample (excludes financial and utility firms). Firms with fewer than 5 calls are excluded. R^2 includes absorbed fixed effects (not within-R^2).